12/08/22

Homework: Figure out how the snowfall in Chennai around 200 years ago is connected to the invention of bicycles.
Answer: 200 years ago, Krakatoa volcano erupted resulting in snowfall all over the world, the year without summer. To counter travel in snowfall, the earliest version of the bicycle was invented.

Seemingly unrelated events or objects could be highly linked to each other.
That's what Network Science deals with.

The Internet is a physical network whereas the Web is a logical network where nodes are pages and edges are the links.

Every time you click a link, it is you, i.e. the user's attention that flows from one page to the other.
So the web is a network of users or people. So it's the world's biggest social network.

Introduction

Network Science deals with models, methods, tools, and mathematical techniques to study and analyze the behaviour of networks. Networks comprise entities represented as nodes (also referred to as vertices), and the relationships among the nodes are denoted by edges (also referred to links).

Networks are everywhere – connecting 'agents' of different types by edges representing their interactions: phone networks connect people through voice, text, or video linkages; electrical networks capture the connectivity between sources of generation and loads which consume the power that flows in the network; biological networks are used to model the nature of interaction between agents representing biological entities such as proteins; social networks model online interactions between social agents – people; and so on.

Network Science today has rapidly emerged as a vast interdisciplinary field of investigation, with tools and techniques drawn from many disciplines, ranging from the basic sciences, such as physics and biology, to the engineering sciences such as electrical engineering, through graph theory and learning in computer science and mathematics, and the social sciences, drawing in topics from microeconomics and game theory.

One of the central goals of Network Science is the study of complex phenomena arising from the interaction of a large number of agents interconnected by a network of linkages. These studies attempt to model and characterize the behaviour of agents located at the nodes, the impact of network structure on such behaviour and their characterization, and the dynamics that may result from changes to the network structure and properties.

Mandates

Very important to know what this system is about…
https://docs.google.com/document/d/1suVvDnzqJkrFv1IywDiEdDaXihngMXh3cAPmoSR0DIE/

Mandate 1: Network Science & Analytics Essentials

Network: A **dynamic** system of interconnected elements, usually an **emergent property** of several independent decision-making.

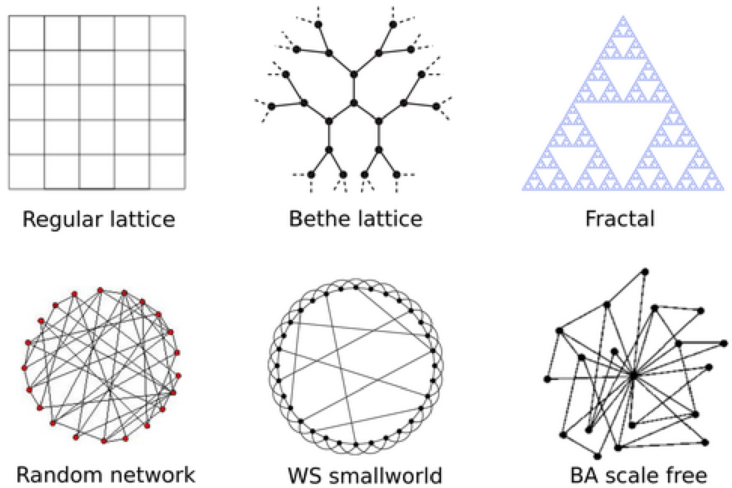
Network Science: Study of networks and their dynamics to derive semantics latent in networks.

Network Dynamics:
Dynamics on a Network (characteristics of flow in a network, like max flow, stress centrality, network flow, etc.)
Dynamics of a Network (characteristics of network evolution)

Network based Reasoning
What can be reasoned from a network, about its resilience, throughput, etc.

Examples of networks,
Networks appear in all domains:
- Protein Pathway Networks
- Ecology Networks
- Airline Networks
- Social Acquaintance Networks
- Transport Networks
- Energy grid networks

The structure of a network provides a number of semantic interpretations of its characteristics. Similarly, the network location of nodes can provide a number of semantic interpretations about its characteristics like: “importance”, “influence”, “impact”, etc.



WS = Watts-Strogatz
BA = Barabasi-Albert

Homework: Randomized algorithm to generate the Sierpinski triangle

Network Analysis, visualization and interpretation, a growing area of interest in several domains,

Strategic decision making
Basic Sciences
Criminology
Policy and Administration
Public Health

Some Network Analysis tools:
Centrifuge: http://centrifugesystems.com/
Cuttlefish: http://cuttlefish.sourceforge.net/
Cytoscape: https://cytoscape.org/
Gephi: https://gephi.org/
igraph: https://igraph.org/
NetLogo: https://ccl.northwestern.edu/netlogo/
NodeXL: https://www.smrfoundation.org/nodexl/
Pajek: http://mrvar.fdv.uni-lj.si/pajek/

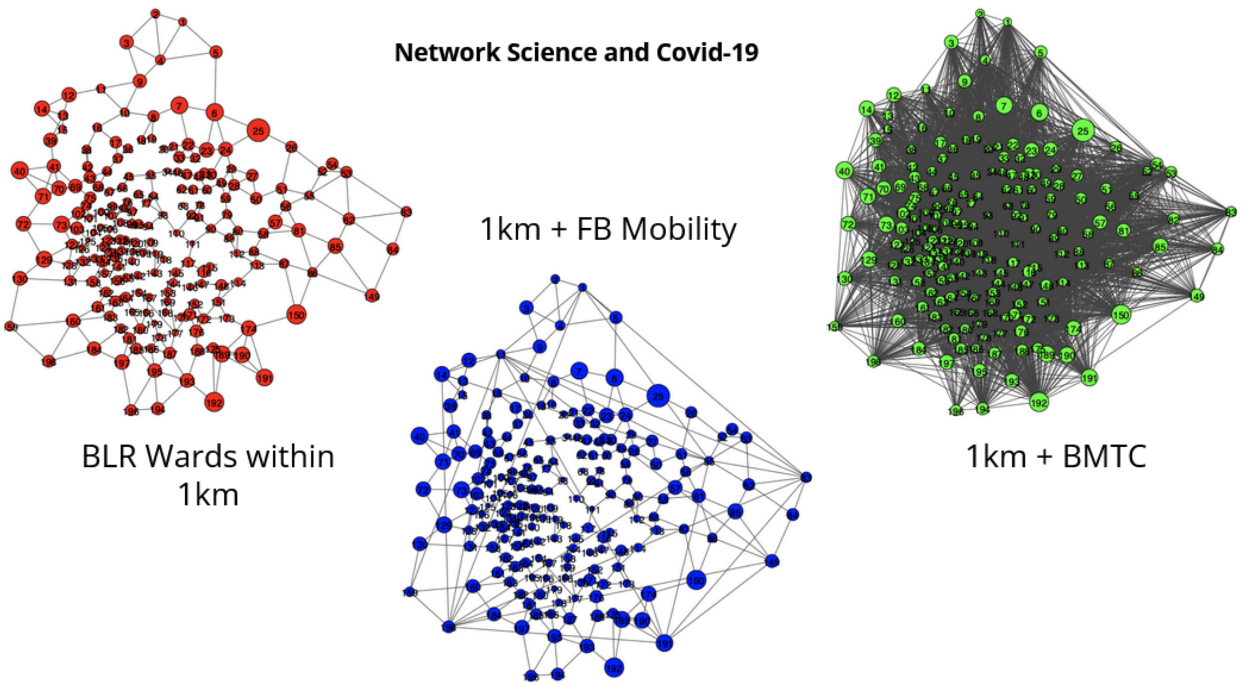
Online Resources:
Online textbook on Network Science by A L Barabasi
http://www.networksciencebook.com/

Networks, Crowds and Markets. Online book by Easley and Kleinberg
http://www.cs.cornell.edu/home/kleinber/networks-book/

Network Science resources

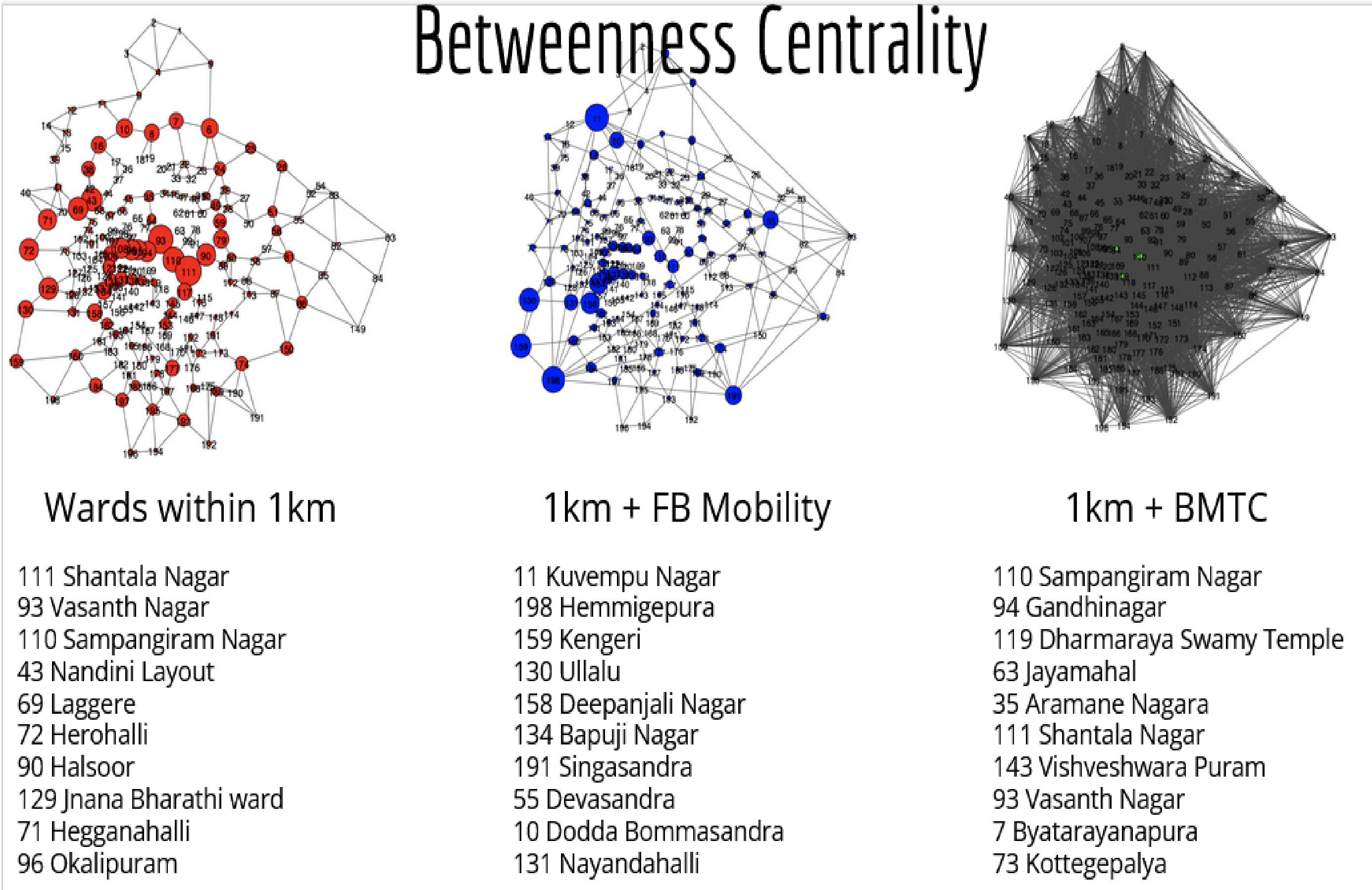
http://network-science.org/

Network Datasets:
UMich Network Data collection. http://www-personal.umich.edu/~mejn/netdata/
SNAP. http://snap.stanford.edu/data/index.html
Dataverse. https://dataverse.harvard.edu/
Pajek datasets. http://vlado.fmf.uni-lj.si/pub/networks/data/
Anonymized Facebook datasets. http://socialnetworks.mpi-sws.org/data-wosn2009.html



Each red node is a ward in Bangalore.
People in a lockdown ward still have freedom to move about for essentials so nearby wards from their own ward can get affected, so diffusion is involved.

On top of this dataset, the mobility of users were tracked using Facebook activity.
And on top of this, BMTC transport network was added.



Computing the Betweenness of each ward in each scenario specified and trying to infer what would be Covid hot zones (the large sized nodes in each graph) in each scenario, i.e. lockdown (grocery runs allowed), private transport allowed, public transport allowed.

Network Science and Graph Theory

Graph Theory forms the underlying formalism for studying networks.
Graphs represent a “snapshot” of a network.
Networks are dynamic and evolve over time, while graphs are static data structures.

The WWW

Global information network that has no parallels in history.
Nothing analogous to the web exists in nature on a global scale.
The web is neither engineered, nor a naturally occurring phenomenon -- It is an **emergent network** formed by billions of independent decisions [think on how a social network and its connections evolve, the app is engineered but the web that’s formed as a result is not; how web pages link to each other to form the web].

Note: Emergent = Source cannot be attributed to one single start point, it is the network as a whole which generates the web.

A Brief History of the WWW

1989

CERN physicist Tim Berners-Lee lays out a proposal for information management called “Mesh”
Original proposal available from http://www.w3.org/History/1989/proposal.html

1990
Berners-Lee changes name to “World Wide Web” while writing code for the Mesh.

Creates three fundamental building blocks:
HTML, URL (later called URI now IRI), HTTP

1990
First web page appears on the Internet

1991
Web available for access to people outside of CERN

1993
WWW code made available for free on a royalty-free basis forever by CERN

1994
Berners-Lee joins MIT to found the World Wide Web Consortium (W3C)

Design principles for WWW adopted by the W3C:
1. Decentralization (no one controls content on the web)

2. Non-discrimination (net neutrality)
No internet routing protocol would look into the data of the packet, so routing is content independent.

3. Bottom-up design (Open source, participatory approach to maintaining web code)

4. Universality (Agnostic to computing platforms or hardware)

5. Consensus (Participatory approach to web standards)

19/08/22

1993
Marc Andreessen from NCSA releases Mosaic – the first graphical browser for the web

1994
Andreessen with two colleagues formed the Mosaic Communications Corporation and released the first commercial web browser called Netscape Navigator.
First International WWW conference is organized at CERN in May 1994

1996—2000
Dot com boom ("Get large or get lost" mantra) and birth of several first generation search engines and e-commerce sites (Yahoo, Excite, Lycos, Altavista, Amazon, …).

Shift in focus from open information online, to secure transactions and e-commerce (SSL, Digital Signatures, etc.).

India among one of the first countries to formulate a regulatory framework for the use of information technology and the Internet (Indian IT Act, 2000: https://en.wikipedia.org/wiki/Information_Technology_Act,_2000)

Stateless protocol = No memory, for HTTP => No memory of previous connections

Stateless is not useful for financial transactions so security is important. So how to communicate while using HTTP which is stateless?
Entire user activity has to be sent back from the server to the client.
Protocols were stateless to save on bandwidth.

How to do this?
RESTful - Representational State Transfer
Session Management - Usage of cookies

How are these different?
Find out on your own :(

1997--1998
Early versions of "social media" sixdegrees.com, moveon.org were born based on popularizing research around social acquaintance networks.

2001
CSCW (Computer Supported Collaborative Work) produces its biggest collaborative authoring ever: Wikis and Wikipedia.

2001—2002

Dot com bust. Major web and Internet companies go bankrupt (Excite, Lycos, Nortel Networks, Worldcom,...).

2002
Web 2.0.
Web reinvents itself as a participatory social medium based on growing popularity of blogs and subscription models (RSS feeds).

2002
SecondLife combines VR and MMORPG (MUD = Multi User Dungeons, Predecessor to MMORPG) to bring virtual worlds to the web.

State transfer happened via text and 3DCG was generated at the client's browser with special browser and proprietary state transfer formats.

2004
Social networking based sites launched in this year: Facebook (in Harvard), Orkut, Ning, Mixi, Piczo, Hyves, Care2

2005
Launch of YouTube. Makes videos a substantial part of web traffic. Catalyzes new technologies for Internet Multimedia protocols and backbone infrastructure.

This was propelled by protocols like IP Multimedia Subsystem (IMS) and technologies like fiber optics.

2006
Launch of Twitter and mainstreaming of microblogging. Facebook dominates social media.

2007
Emergence of Smartphones with the launch of iPhone by Apple, and the opening of Android ecosystem by Google.

2008 -- 2010
Web Socket protocol enables multi-modal interaction on the WWW. Several businesses built around social media: Pinterest, Spotify, Groupon, Instagram, Foursquare.

Web Sockets are stateful. Bandwidth is no longer a concern.

2009
Bitcoin becomes the first cryptocurrency that is traded with real currency. Uses underlying technologies of distributed ledgering, and "proof-of-X" as a family of consensus mechanisms.

2013--till date

Convergence of mobile data, GPS, MEMS, RFID, wearable computing, etc. to start the era of web-enabled smart devices, or the Internet of Things (IoT).
Increasing use of social media in political campaigns.
Increasing concerns about use of individual data by commercial and political interests.

2014
Infamous Facebook experiment to mass manipulate emotions.

Web 3.0: Crypto, Blockchain, Smart Contracts, NFTs, allied technologies, etc.

Focus of each iteration of the web,
Web 1.0: Attention
Web 2.0: Participation
Web 3.0: Ownership (tech like blockchain, i.e. ledgering technology)

2018
Sir Tim Berners-Lee issues a global declaration for a "Bill of Rights" or "Magna Carta" for the web, amid increasing concerns of erosion of founding principles of the web. Calls for development of architectures for "re-decentralization" of the web. Proposes an architecture called "SOLID" for giving users control of their own data.
https://www.commondreams.org/news/2018/11/06/magna-carta-web-read-tim-berners-lees-global-declaration-humane-and-democratic

https://solidproject.org/

2021
Idea of Web3 gains popularity-- bringing together technologies of blockchain, cryptocurrency, NFTs and smart contracts, to build a "token" based decentralized architecture for establishing ownership and privileges in online transactions.

Models of the Web

The web is unlike any other technology developed so far -- it is neither engineered, nor natural. Unlike say cars or washing machines, there is only one web.
Is the web a "technology" or a "tool" that we use or is it something else?

Notable paradigms of the Web considered by researchers:
- Very large database
- Digital library
- A cognitive extension of ourselves
- Participatory socio-cognitive space

Web as a Database

Early approaches (mid '90s) to model the Web.
Focused on the “semi-structured” nature of the Web and as a special case of managing structured (RDBMS) databases.
Research objectives: structured and rich query semantics.

```
select
  source_content as DOCUMENT,
  source_title as TITLE,
  source_url as URL
from
  crawl of http://news.google.com
  to depth 2
  following if url_host(URL) not matching 'google'
join where URL not matching 'google'
select
  URL,
  TITLE,
  summarize(clean(ARTICLE_BODY), 3) as SUMMARY
from
  articles
within
  inline DOCUMENT
```

An example WebQL query

Source:
http://en.wikipedia.org/wiki/WebQL

Even this assumption is too structured for the web and that’s why this model of the Web faded away eventually.

Web as a Digital Library

Indexing or Inverted Indexing which is used in an actual physical library began to be used here. Find a book by author name and book title, give these details in the index card and it will tell you the exact location of the book in the library. Very analogous to how a keyword-based search engine works.

Major shifts,
Strict notions of “query” to looser notions of “retrieval” and “relevance”.
Strict notions of “schema” to looser notions of “ontology” [Ontology includes the questions of how entities are grouped into basic categories and which of these entities exist on the most fundamental level].
Emphasis still on retrieving information.
The Web is still seen as a passive repository of information.

Web as an extension of our mind

Rooted in Vannevar Bush's interpretation of hypertext reflecting the way information is organized in human brains.

Focus on interpreting hyperlinks, rather than (just) data on web pages
Hyperlink as a(n):
1. Relevance indicator
A hyperlink from page A to page B probably means that page B is somehow relevant to the contents of page A.
Also, where a particular hyperlink is present on a particular web page can indicate how relevant the hyperlinked web page is to the current web page.

2. Endorsement
A hyperlink on a web page is like an implicit encouragement to the user to click on the hyperlink and visit the hyperlinked web page. So the current web page is basically “endorsing” the hyperlinked web page.

3. Attention pathway
A hyperlink is a pathway for your attention to flow from page A to page B.

That’s why this model is called the web as an extension of your mind because links are part of your mind only because you are making the conscious decision of clicking on the hyperlink and transferring your attention. Also this is how our memories are stored as well according to Vannevar Bush atleast.

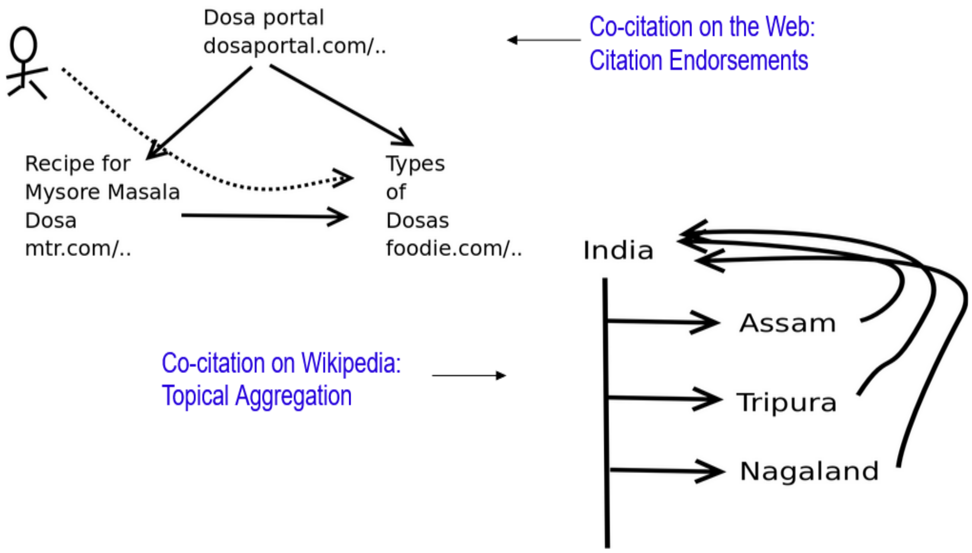
How memory links are established => Cells that wire together, fire together.

26/08/22

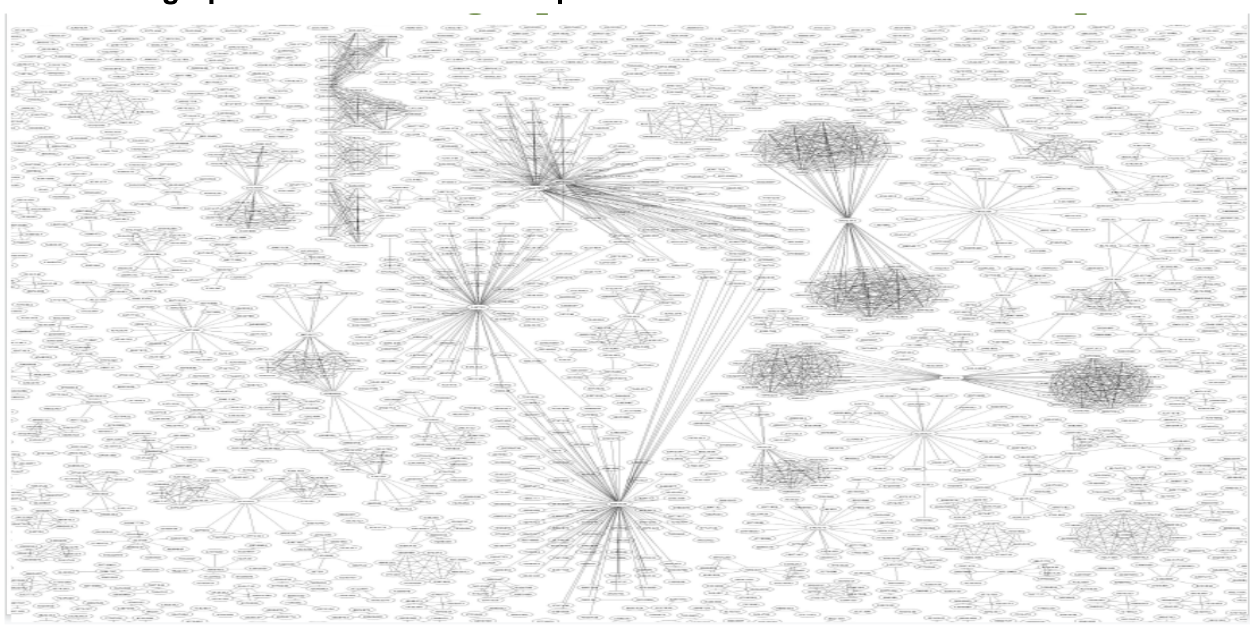
Sidenote: Co-citations and Endorsed Co-citations

Consider 3 pages A, B and C. Pages A and B are highly co-cited, if there are many pages like C which link to both A and B.
High co-citations typically mean that A and B are very relevant to topics that pages like C cover.

Why is the Hyperlink Distance for the Wikipedia graph mostly 2 and not 1 like for the Web graph?
On Wikipedia, citations get reorganized into hierarchical topics. So highly co-cited topics like “Assam”, “Tripura” get put under the umbrella of “India”, so Assam doesn’t directly link to Tripura but instead one has to go from Assam to India to Tripura, thereby making the hyperlink distance 2. These topics put under a single umbrella like “Tripura”, “Assam”, etc. are called **Semantic Siblings** because they can be used in the same context and replaced with one another without much issue syntax-wise.



Co-citation graph from a web crawl sample



Look at the fun patterns forming in this web.

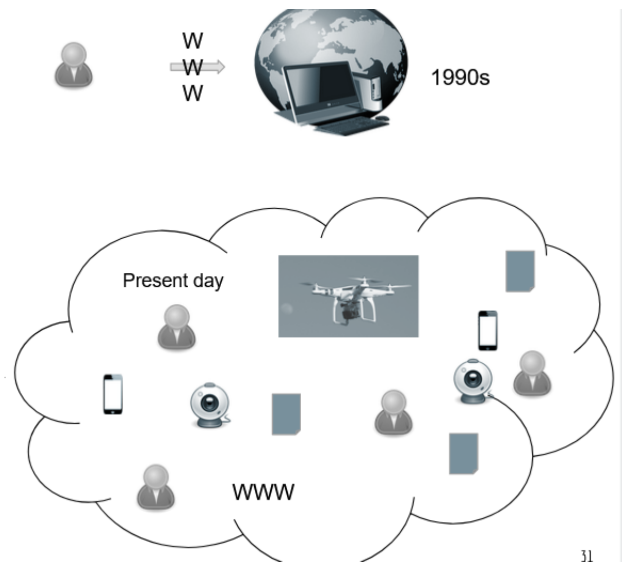
Attention Backbone

Citations represent flow of attention in a hyperlinked repository.
Endorsed citations distinguish between “highways” and “bylanes” in the attention flow network.
The hyperlink graph formed by only highly endorsed hyperlinks called the Endorsed Citation Graph (ECG), forms the “Attention backbone” of the hyperlink repository.
ECG for web crawl was shown to be made of disconnected components with a skewed distribution of component sizes.

Endorsed Co-citation
Pages like A and B are highly co-cited but the linking pages don't refer to other pages even if A and B themselves have other pages linked.
It means that A and B are pages that are highly relevant to the topics that pages like C are talking about and it is sufficient enough for the authors.
This means that pages A and B are endorsed co-citations.

Web as a socio-cognitive space

A **logical space** subsuming our existence. Proximity based on whom you follow => are "linked" to.
Humans as "participants" rather than "users".
The Web uses us as much as we use the web!
Perhaps for the first time, tech research is as much about how to avoid getting exploited by technology, as about how to exploit the technology!



Notice how you are actually a part of the Web in the Present Day section.
Privacy is obviously a concern but the real danger is freedom of thought and independent thinking and how we can be manipulated by the Web to do what we might not have done otherwise.

So in reality, humans are becoming "Components" of the web instead of "Participants".
Components = Involuntary, Participants = Voluntary

Characteristic elements of the socio-cognitive space paradigm:
- Crowdsourcing
- Participatory authoring (Ex: Wikis)
- Social media shares
- Social Recommendations
- Push-based notifications

- Location-based services

Now that Users are also part of the Web, it is no longer the User’s Attention that’s flowing from a link to another page. It is the User’s Opinions that are flowing through the Web.

Social Learning

Attention networks bring about things like Page Rank, i.e. which pages are important over others.

What do Opinion networks bring about?
Belief Revision.
Revising the beliefs of users regarding some topics.

Study of how agents “learn” (undergo semantic changes in their worldview) in a social setting
Models the dynamics of opinions-- their spread, adoption, interplay, network impact, etc.

Tool vs Space
Tool = Extends your capabilities
Space = Subsumes you, you exist in a space

Opinion comprises of,
Abstraction / Idea
Expression / Sentiment

Idea = What the opinion is talking about
Sentiment = How the opinion is presented

Networks on the web

Hyperlinks, Citations, Wikilinks Networks
Relevance indicator
Endorsement
Attention pathway

Social Media Networks
Follower networks
Acquaintance networks
Retweet graphs
Mention graphs

Graph Theory Fundamentals

General form of a graph: G = (V, E)

V: Set of nodes or vertices
E: Set of edges or links

Undirected graph: $E \subseteq V^2$
Undirected edge: {1, 2}

Directed graph: $E \subseteq V \times V$
Directed edge = Ordered Pair = (1, 2)

Terms to define
Connectivity = How many nodes/edges have to be removed for the graph to become disconnected

Free tree = Tree with no root

Diameter = Degree of separation
[tree with minimum diameter = 2 => Star network => Single Point of Failure]

Efficiency = Lower the diameter, higher the efficiency, because to go from one node to another, you require less number of edges.
And each edge brings with it various factors like delay, distortion, etc. that reduces the efficiency.

Resilience = How resilient a graph is to being disconnected
How can resilience be tested,
- Random failures
- Targeted attacks

Neighbours of a node v in a graph:

$$\Gamma_v = \{u \mid \{u, v\} \in E\}$$

Degree of a node:

$$d_v = |\Gamma_v|$$

Neighbours in a digraph:

$$\Gamma_v^i = \{u \mid (u, v) \in E)\}$$
$$\Gamma_v^o = \{u \mid (v, u) \in E)\}$$

Indegree and Outdegree:

$$d_v^i = |\Gamma_v^i|$$
$$d_v^o = |\Gamma_v^o|$$

Undirected graphs cannot have total degree sum as odd.
Regular graphs => r-regular

What does a regular graph look like when r = 0, 1, …?
r = 2 => Smallest r value to get connected graph (ring structure).

Need r+1 nodes to build r-regular graph

Complete graphs, its properties regarding diameter, connectivity, etc.

02/09/22

Average degree:

$$\langle d \rangle = \sum_{\forall v \in V} \frac{d_v}{|V|}$$

Degree distribution:

$$p_k = \frac{N_k}{N}$$

Distribution of probabilities of node degrees. N_k is the number of nodes with degree k, and N = |V|.

Weighted graphs:

G = (V, E, w)

w: E → Re

Signed graphs:

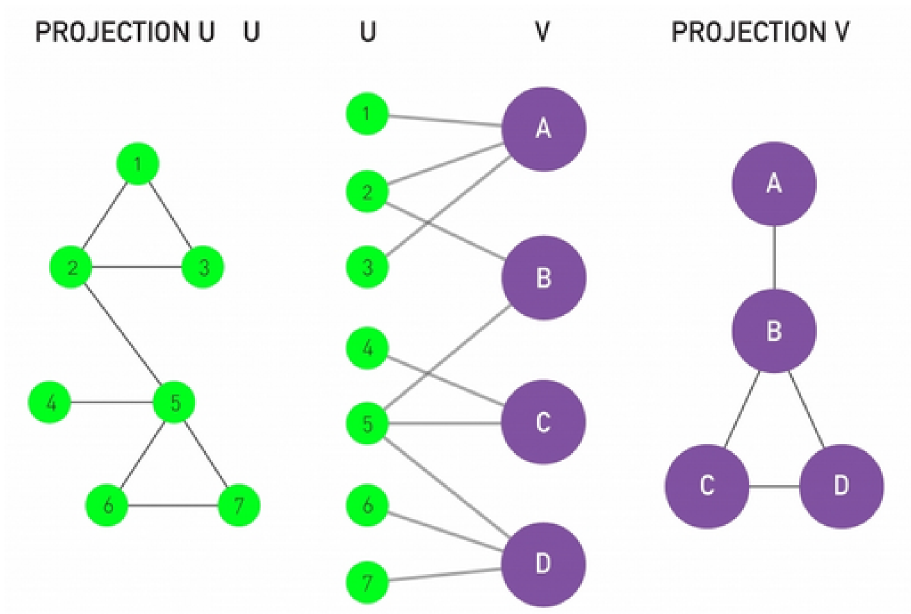
G = (V, E, s)

s: E → {-1, 0, +1}

Bipartite Graphs and Projections

A graph G = (V,E) is called a bipartite graph (similarly k-partite graph) if there is a labeling function l: V → L to each node that assigns a label from one of two labels to each node.

for any edge e ∈ E:
label(source(e)) ≠ label(target(e))



Information Entropy
Amount of bits required to encode any message is called Information Entropy.
Given by the formula :-

$$H = -\sum_{i=1}^{n} p_i \cdot \log_2(p_i)$$

Interesting degree distributions:

- Regular graphs have a unimodal degree distribution
- Random graphs have Poisson or Gaussian distributions
- Epidemic networks have an exponential degree distribution

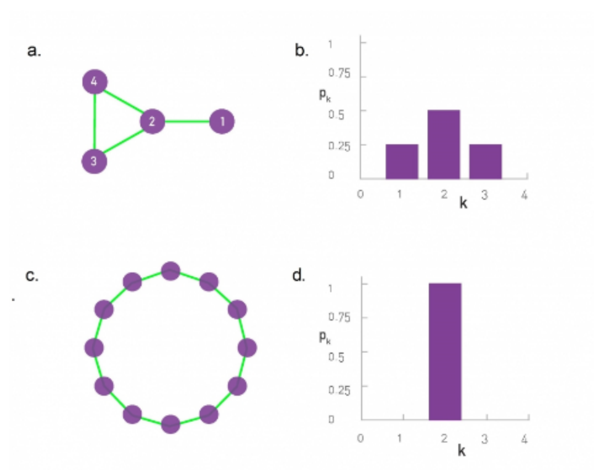
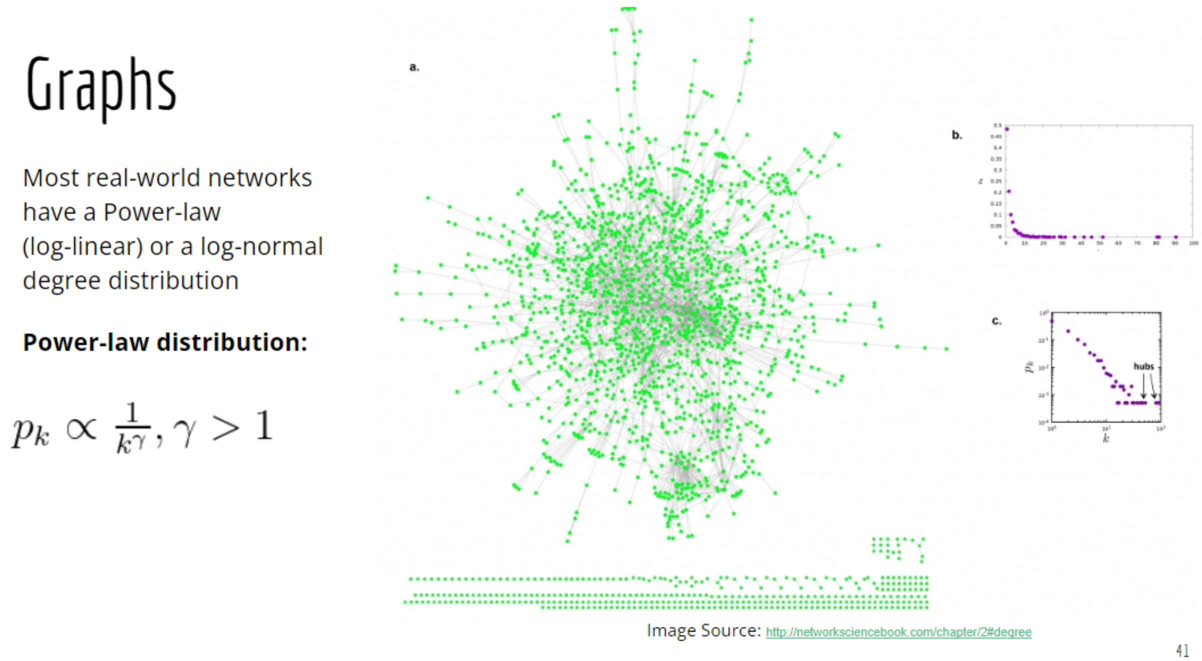


Image Source: http://networksciencebook.com/chapter/2#degree

Power Law



Power law distribution talks about the probability of finding a node with degree k is proportional to $1/k^{\gamma}$. Power law network is also called an **Infinite Variance** network.

The graph above shows that 80% of values occur in the first 20% of data, also known as the **Pareto Principle** also called the **80-20 rule**.

Pareto Principle

The 80-20 rule, also known as the Pareto Principle, is a familiar saying that asserts that 80% of outcomes (or outputs) result from 20% of all causes (or inputs) for any given event.

In a power law distribution, there are a small number of nodes with a very high degree and a large number of nodes with a small degree. The very high degree nodes are called the **Hubs.**

Note: Read about power law and find out data about airports and airlines regarding power law.

Paths / Walks
Path is a structural construct whereas Walk is a traversal construct. A walk can have repeated edges but a Path cannot.

Note: Plot a visualization of BMTC dataset to verify they employ a Hamiltonian Cycle.

Girth Of A Graph
In graph theory, the girth of an undirected graph is the length of a shortest cycle contained in the graph.

What is the minimum number of edges for n nodes where we can guarantee a triangle?

For **floor(n^2/4) +1** edges, there will definitely be a triangle present.
Consider a bipartite core (bipartite graph that is complete) with n vertices and n/2 vertices in both halves, so there are $(n/2)^2$ edges.

We for sure know that there's not a triangle in this graph because that's how bipartite cores work. And when you add one more edge in this bipartite core, it will no longer be a bipartite graph and there is guaranteed to be a triangle, so we get, $(n/2)^2$+1 edges.

Note: Read about (floor(n^2/4 +1) + 1 - k * m) / n^2, where k = no. of triangles and m is the number of edges.

Triadic Closure
What Facebook / Linkedin use to suggest friends / connections in **"People You May Know".**

The limit to form 1 triangle is, $(n/2)^2$ + 1 edges and say you have a graph with m edges and k triangles.

If you have k triangles even with a small number of edges it means that this graph is prone to form triangles and there's strong connectivity which is what forms the basis for Triadic Closures.

09/09/22

Hamiltonian Graph
Graph that contains a Hamiltonian Cycle.
Minimum connectivity is 2
Diameter can be at most n/2
Atleast n edges

If you try to optimize the Efficiency, Cost and Resilience of a Network, then you'd find that the network resolves to a Hamiltonian Graph.

Hamiltonian Graphs are also called Circular Skip Lists (CSLs) because Hamiltonian introduces a Circular Linked List type of structures and then you can have skips from a node to another node that is not its cyclic neighbour.

Signed Graphs
Undirected graph that contains +ve (consonance) and -ve (dissonance) signs on the edges.
Also called as Graph with -arity.
These signs could indicate many things such as sentiment on a particular topic in an online forum.

+ve edge could imply greater strength of association, greater trust and lower bookkeeping cost (in terms of discussion forum, one can speak freely and there's less worry of making "mistakes", in terms of relations between countries, this could mean the amount of money spent in border security, if relations are good, then very less money required for border security).
You can think of the opposite for -ve edges.

Signed Graphs link to **Network Stability** as negative edges indicate Network Instability.

To make a Network Stable you have to cut off all the negative edges (which may not be feasible depending on what kind of network and what kind of situation your network is in) and you could get situations like A likes B, C likes B but A doesn't like C resulting in an unstable graph which cannot be easily made stable.

Spectral Graph Theory
Using Linear Algebra to talk about Graphs and Networks.

Adjacency Matrix
[Insert info]
Undirected graphs have a symmetric adjacency matrix.
Directed graphs with a non-empty diagonal have self loops.

If A is the adjacency matrix,

What does $A^2 = A \times A$ represent?
What does this $A^2_{i,j} = k > 0$ mean?
This means that there's a path from i to j of length 2.
The value k would indicate the number of such paths of lengths 2 that are available from i to j.

Similarly, A^k would indicate paths of length k between any i and j

What does A* or A^∞ indicate?
Transitive Closure.
Repeating the process of exponentiation of the matrix till all pairs of vertices become connected, i.e. each entry in the matrix is non-zero.

"The reachability matrix is called the transitive closure of a graph."

Linear Transformations
$A * x = b$
Where A is a matrix and x and b are column vectors.

Depending on size of m and n, you can dimensionality reduction, dimensionality increase (kernel trick in SVMs).

x can also be looked at as the State of the Network if A is the adjacency matrix of the network.
x tells us the value of some parameter for each node in the network.
And b will tell us the state of the network wrt some other domain.

Eigenvectors

Given an adjacency matrix AG the eigenvector of this graph is the vector satisfying the equation:

$$A\vec{v} = \lambda\vec{v}$$

The eigenvector is also called the self-vector because it represents the point of invariance.

In terms of A * x = b,
If x is the eigenvector, then b will have the same distribution as x just scaled by the eigenvalue and this distribution is an invariant of the graph.

In directed graphs, when edges represent some form of “flow” (influence, attention), the eigenvector of the graph represents the dominant invariant distribution of the element under flow.

The term λ is the eigenvalue, representing a scaling factor for the invariant distribution.

The eigenvector with the highest value of λ is called the principal eigenvector, while the rest are the non-principal eigenvectors.

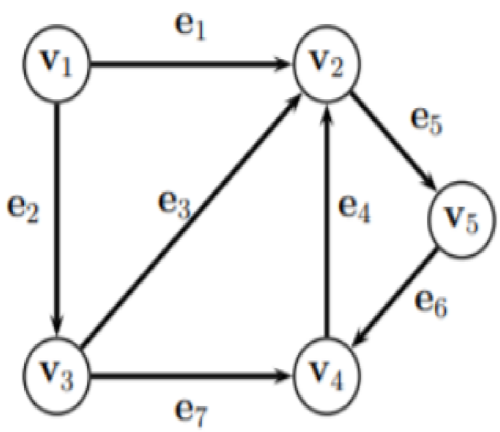
Adjacency matrix is also sometimes called the “diffusion operator” of the graph.

Incidence Matrix

For directed graphs, adjacency coupled with direction, gives rise to the incidence matrix.

Given a digraph G = (V,E) where |V| = m and |E| = n, the incidence matrix B_{mxn} is defined as:
B[i,j] = +1 if source(e_j) = v_i
B[i,j] = -1 if target(e_j) = v_i
B[i,j] = 0, otherwise

Example,



$$B = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & -1 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}$$

[Rows are vertices and columns are edges]

Graph Laplacian

Analogue of differential operators in calculus. The Laplacian of a graph G is given as:

$$L = BB^T = D - A$$

Here, DG is the degree matrix of G where:

D[i,j] = d_i when i = j

D[i,j] = 0, when i ≠ j

A_G is the adjacency matrix of G

The diagonal elements would indicate the degree of the node and the non-diagonal elements will be -1 if there is an edge between i and j otherwise it would be 0.

With weighted graphs, A is replaced by W, indicating the weights of each edge, and degree defined as the sum of weights of all incoming edges.

The Laplacian of any matrix is symmetric and positive semidefinite.

Let X: V → R_e be a function allocating some quantitative metric (like attention, influence, wealth, etc.) to each node. Then:

$$x^T L x$$

is called the graph derivative.

$x^T L = x^T * (D - A)$
x represents the distribution of some parameter over the network, i.e. state of the network.

The first row of the result can be represented as,

$$d(v_1) \cdot x(v_1) - \sum_{\forall u \in \Gamma(v_o)} x(u)$$

This computes the total amount of x that's being given to v_1 **by considering its degree** and then subtracts it from the amount of x being given to its neighbours.

So it gives the relative/differential standing of a node from its neighbours.

Let G = (V, E) be an undirected graph, and let X: V → Re be a function that assigns a quantitative metric to each node in the graph.

The term:

$$x^T L$$

corresponds to the differential value of each node, with respect to its neighbours

$$\nabla_{v_i} = \sum_{j \sim i} (x_i - x_j)$$

$$d_i \cdot x_i - \sum_{j \sim i} x_j = \sum_{j \sim i} (x_i - x_j)$$

The graph derivative is defined as:

$$\nabla_G = x^T L x = \sum_{i=1}^{|V|} x_i \nabla_{v_i}$$

We can see that if all nodes have the same value, the graph derivative is 0.

A positive graph derivative indicates that there's atleast some nodes which has higher values (for x) as compared to its neighbourhood.

What would be the case for negative graph derivatives?

What would be the case for the eigenvector having a positive graph derivative?
It would represent the same distribution as the graph but scaled up by some value.
And it would be scaled by some negative value if the graph derivative is negative.

Graphs & Rank of Matrix

Rank of matrix = Number of linearly independent column/row vectors of the matrix
It represents the basis of the vector space of the matrix.

What would rank indicate for the adjacency matrix of a graph?
If the rank of the matrix is low, it means that a low number of independent columns/rows of the graph are required to obtain all the other columns/rows of the graph. And each column/row represents each node's adjacency, i.e. the neighbourhood of a node.

So if the rank is low, it means that many nodes have similar neighbourhood and vice-versa if the rank is high.

Embeddings
Representing Graph in n-dimensional space
Representing in 2-dimensional space is called Layout where each vertex is given some (x, y) coordinate.

Convolutions
Similar to the Convolutions in Neural Networks.
In graphs, convolutions refer to,
Representing the entire neighbourhood of a given node by a single virtual node which is computed via a convolution of some operation over the neighbourhood and then moving on to another node in the neighbourhood and repeating the same..

Analytics Fundamentals

Analytics
Observing a system to make inferences about its constituents, structure and phenomena, in order to obtain actionable intelligence and/or to predict its future dynamics.

Paradigms of Analytics:
Confirmatory Analytics
- Statistical Hypothesis testing

Exploratory Analytics
- Visual data mining

Generative Analytics
- Stochastic Modeling

16/09/22

Some Definitions

Population
The set of data points to study.

Sample Space
A subset of the population.
The sample has to be representative of the population and its characteristics.
So there are sampling strategies to manage this accurately.
Probability & Likelihood

Probability
What is the possibility that an observable event happens given a causal event has happened?
Causal Event = Model (represented by theta) = Latent variables, that can potentially cause the observable events to happen.
Example, what is the probability that North Bangalore is flooded given that we received 50mm rain today?
50mm rain is part of the model in this case.

Likelihood
We know the event that has happened and we are trying to find out the “likelihood” that it was caused by some model/causal event.

Example, North Bangalore is flooded, what is the likelihood that it was caused by 50mm rainfall?

Confirmatory Analytics

Begin with a hypothesis about a specific construct.
Collect data to test the hypothesis.
Perform statistical inference on the data.
Confirm/fail to confirm the hypothesis.

Statistical Inference

Core element of confirmatory analytics.

Process of deducing properties of one or more latent causal parameters, based on observed distributions.

Paradigms of statistical inference:

- Frequentist inference => Depends on the frequency of the relevant data variables.
- Bayesian inference => Use the prior knowledge
- Information theoretic inference
    - MDL

Frequentist Inference

Estimates values of true distribution of data, by (repeated) sampling of data and measuring sample distributions.
No prior domain knowledge necessary.

Example:
Over the last N observations of flight delays at the airport, n1 were due to technical snags, n2 were due to fog, and n3 were due to VIP movement.

Given that flights are delayed at the airport today, what is the probability that it is due to VIP movement?

$$L(VIP|delay) = \frac{n_3}{N}$$

Bayesian Inference

Combines prior knowledge about population distributions, with evidence obtained from data, to compute a posterior estimate.
Based on the Bayes theorem of conditional probabilities:

$$P(H \mid E) = \frac{P(E \mid H) \cdot P(H)}{P(E)}$$

H: Hypothesis, E: Evidence
P(E|H): Likelihood of evidence given hypothesis
P(H): Prior probability of H
P(E): Marginal likelihood or “model evidence”

Extending the previous example. Let prior probabilities of flight delays due to various factors be represented as: P(snag), P(fog), and P(VIP) respectively.

Given that flights are delayed, the probability that the delays are due to VIP movement, is given by:

$$P(VIP|delay) = \frac{P(delay|VIP)P(VIP)}{\sum_{cause \in \{snag, fog, VIP\}} P(delay|cause)P(cause)}$$

Replace all the P’s by L’s because this is Likelihood and not Probability.

Statistical Hypothesis Testing

Also a core element of confirmatory analytics. Based on “testing” whether data supports a stated hypothesis (called Ha or H1 or alternate hypothesis).

Alternate hypothesis is contrasted against a “null” or “control” or “default” hypothesis H0 which states the status quo or the negation of the alternate hypothesis.

For example, testing whether Covid vaccine is useful.
How is testing done?
Randomized Control Trials
Get random samples of people, make two groups,
Control Group => Given placebo
Test Group => Given the actual vaccine
And test each group’s statistics to fight Covid.

Example:
Alternate Hypothesis, H_a: Use of social media is detrimental for studies, as reflected in test scores
Null Hypothesis, H_0: There is no evidence to say that use of social media is detrimental for studies, as reflected by test scores

Test Statistics

A sample data set is collected based on the stated alternate hypothesis from which a test statistic is drawn.

Common test statistics:
- Z-tests: useful for comparing means under i.i.d.
- T-tests: useful for small samples where population variance is unknown
- Chi-square tests: useful for testing variance, independence and goodness of fit
- F-tests: Useful for testing goodness of fit

Paired t-test. Collect two samples of n data elements.
Sample 1: test scores of students not using social media
Sample 2: test scores of students using social media

Compare differences using the t-test:

$$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}$$

$\bar{d}$ = Sample mean of differences
d_0 = Hypothesized mean of differences
s_d = Standard deviation of differences

Differences = Difference between the target values of the Test and the Control Groups, example, difference between the test scores of the students that used social media vs students that didn't use social media.

Significance Level

A test measurement is said to be statistically significant, if it is not likely to be the outcome of pure chance.

To determine this, the p-value of the measurement is first calculated by projecting it onto a probability distribution depicting the assumption of the class of the system represented by the trial.

A hypothesis test is said to be statistically significant, if its p-value is above a predetermined significance level α (usually set to 5% or 0.05).

Types of Errors

Rejecting the null hypothesis based on p value may introduce Type I errors (where the alternate hypothesis is false, but accepted) => False Positive

Retaining the null (rejecting the alternate) may introduce Type II errors (where the alternate is true, but rejected due to lack of evidence) => False Negative

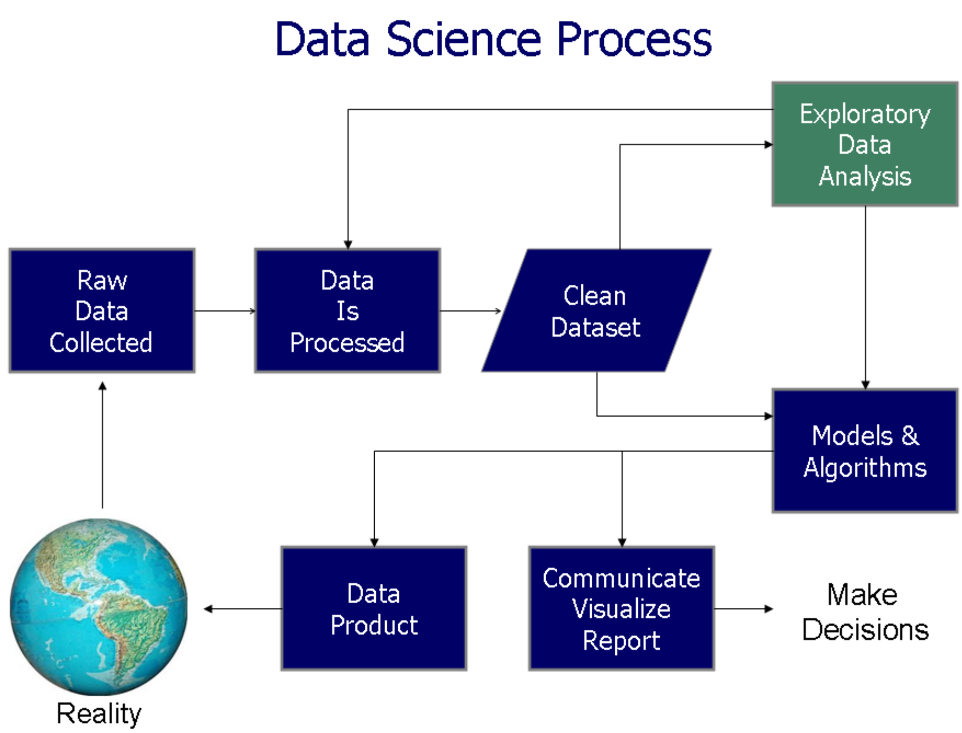
		Decision	
		Retain the null	**Reject the null**
Truth in the population	True	CORRECT $1-\alpha$	TYPE I ERROR α
	False	TYPE II ERROR β	CORRECT $1-\beta$ POWER

α value represents the largest probability of committing a Type I error. Type II errors when an observation yields a test statistic whose likelihood is falsely below the significance level. Such errors are also called β errors.

Exploratory Analytics

Exploration of datasets usually with the help of visualization techniques, to aid in statistical model building or data cleaning.
No confirmation of hypotheses.

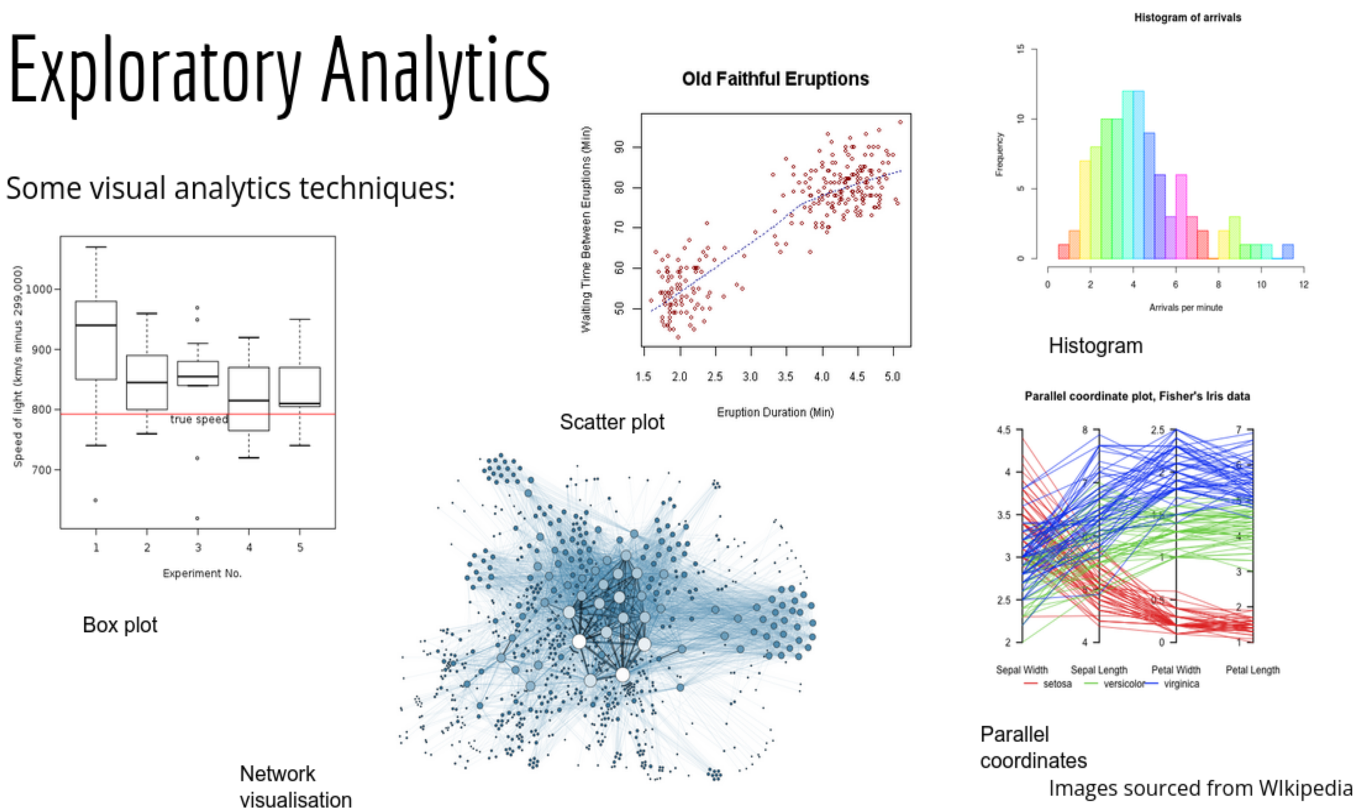
Some information theoretic techniques used to maximize information content in projection:
- Dimensionality reduction
- Multidimensional scaling
- Principal Component Analysis
- Exploratory Factor Analysis
- Manifold learning



Interesting-ness Criteria
Things that we look for via EDA,
Frequency = Frequency patterns in the data
Rarity = Relates to outlier detection
Periodicity
Burstiness
Invariance

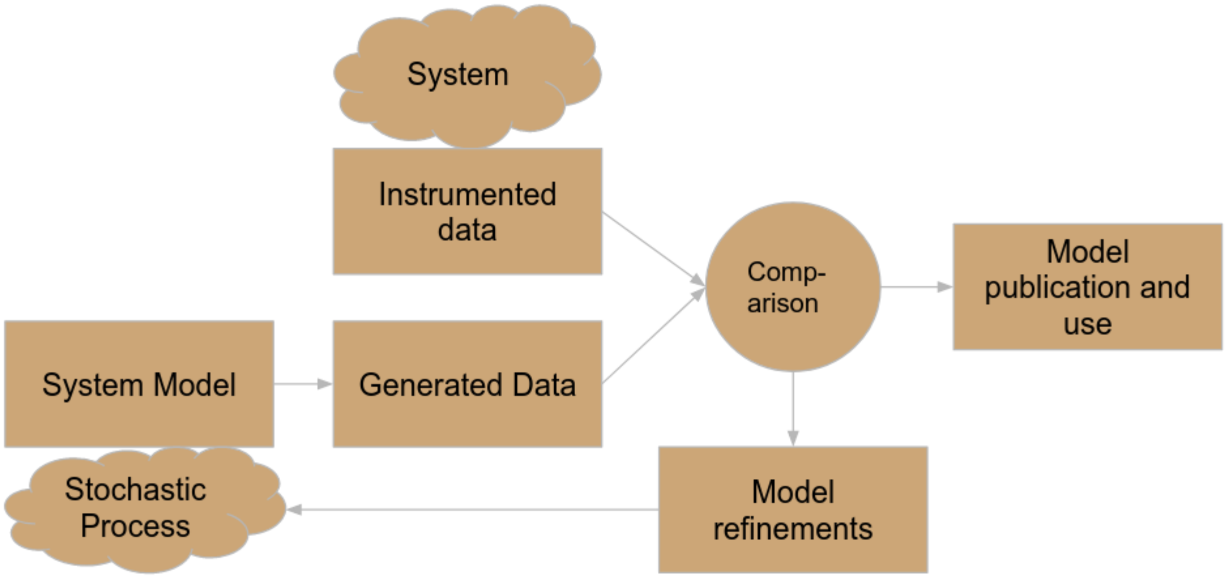
Visual Analytics

Core component of Exploratory Analytics.



Network Visualization = Spring layout of graph (higher value = larger nodes, etc.)
Parallel coordinates = Used for visualizing multi-dimensional vectors

Generative Analytics



The goal is to not test a hypothesis, instead we build an entire system (theory) that supports observations.

Concerns the development of abstract system models to explain observational data from a system

System models are simulated to generate synthetic data that are compared with actual data

[This comparison tells us how well the System Model is able to explain the patterns presented by the actual data which in turn tells us the correctness of the model.]

System models are based on assumptions about underlying stochastic phenomena characterizing the system.

Stochastic process: Formal models to describe different classes of stochastic effects of system dynamics

Stochastic Phenomena

Additive Stochastic Effects

Aggregate property of the system represented by a linear weighted sum of values of individual random variables.
Ex: Aggregate vehicular pollution in a city, represented as a sum of vehicular pollution from individual vehicles.

Multiplicative Stochastic Effects

Aggregate property of the system represented by a weighted product of values of individual random variables.
Usually seen in situations where there's a sequence of events. Exponential growth comes to mind.
Ex: Number of bacteria in a petri-dish at a given time is represented by the product of the population at previous time step and growth rate of individual bacteria.

Logistic Equation/Map
Linked to chaos theory.
https://en.wikipedia.org/wiki/Logistic_map

$x_{t+1} = r \cdot x_t \cdot (1 - x_t)$
$r = \text{Rate of growth}$
x_t that represents the ratio of existing population to the maximum possible population

Homework: Plot values of this equation for some iterations for some value of r like 1.5 and so on, and some initial value of x_t where x_t is number of fish in the pond.

Correlated Stochastic Effects

Aggregate property of the system is a function of **generalized dependencies (i.e. no clear pattern)** between random variables. Independent variables can interact arbitrarily.
Ex: Average speed of vehicles in a road network, a function of individual speeds of vehicles and interferences between vehicles.

Central Limit Theorem

An aggregate property of a set of random variables under certain conditions, converges to a stable "attractor" distribution.

For additive phenomena, the attractor distribution is a Gaussian whereas for multiplicative distribution it is an Exponential distribution.

Going by the definition, correlated phenomena shouldn't have any underlying distribution because the patterns are not known. But many empirical studies have shown that the following distributions appear in networks with correlated phenomena like power law, Dirichlet and log-linear.

If the random variables were i.i.d. with finite variance, the attractor distribution is usually a Gaussian or "normal" distribution.

Generalized CLT addresses several variants of types of random variables and their conditions, leading to different attractor distributions.

Stochastic Processes

i.i.d. Process

Represents a system comprising of several independent, and identically distributed random variables

Aggregate property, an addition of values from individual random variables

Central limit theorem: Overall distribution of aggregate values approximates to a Gaussian (normal) distribution when individual distributions have finite expected value and finite variance

Examples:
A set of outcomes from a fair die
Distribution of white noise amplitudes
Plot of Math.random() values for large number of trials

Bernoulli Process

A discrete-time stochastic process representing a finite or infinite series i.i.d. random variables X_1, X_2, …, that have only two outcomes: 0 or 1.
For a Bernoulli process of length n, if the probability of X=1 is p, then the probability of finding k 1s in the process would be:

$$P[num_1 = k] = \binom{n}{k} p^k (1-p)^{n-k}$$

Example:
Following the sudden announcement of a new policy (like demonetization), the first few reactions on social media (for/against) tend to be independent of one another, and can be modeled as a Bernoulli process.

Martingales

A martingale or a “gambler’s process” represents a sequence of random trials, where knowledge of earlier outcomes has no bearing on the subsequent outcome.

Formally:

$$E[X_n | X_{n-1}, X_{n-2}, \ldots, X_1] = X_{n-1}$$

This equation means that the Expected Value for X_n is the same as that for X_{n-1}

[

Gambler’s Fallacy:
I’ve lost 15 times already, now my luck will change and I will win.
The fallacy is that the probability is the same as the first time and it has never changed.
]

Example:
Given any tweet, it is likely to be either retweeted or ignored, by the next reader. Given this, the expected number of retweets of a given tweet #t over time, can be modeled as a martingale:

$$E[numRT_{t+1}(h)] = numRT_t(h)$$

Random Walk

A random walk represents a stochastic process involving a series of random steps taken in a mathematical space (a mathematical set with a concept of adjacency).

The step taken by the process at step k, is based on its “location” at step k-1 and that is dependent on the location at k-2 and so on.

So there’s a notion of dependency between steps of the process which was not the case in all the processes till now.

Example: Trajectory of a random user browsing some part of the web can be approximated as a random walk on the web graph.

Chinese Restaurant Process

Models stochastic distributions influenced by a base distribution H.

Values obtained by random variable X_t at time t influenced by base distribution H_t at time t.

Base distribution H_{t+1} at time t+1 is dependent on the value taken by a random variable X_t at time t.

Models non-linear interdependencies between stochastic process and base distribution.

Imagine a Chinese restaurant with an infinite number of circular tables with infinite capacity:
Customer 1 chooses a table uniformly at random to sit on.

The k-th customer chooses a table with a probability proportional to the number of people already seated at the table, i.e. base distribution H_{t+1} is dependent on the number of people already seated => X_t

Also called **generalized preferential attachment process.**

This process is very similar to the Polya's Urn model.

Indian Buffet Process

A small variant on the Chinese restaurant process. Given a buffet spread of n tables, a user creates an assortment of dishes on their plate by choosing dishes from table k with the probability,

$$\Pr(k) = \frac{b_k + 1}{n}$$

where b_k is the number of users who have already chosen the dish at table k.

Useful for modeling popularity of online groups on social media. Membership to a group is proportional to the popularity of the group, which in turn is based on the number of people who have already chosen that group.

Other similar processes:

Dirichlet Process

D(H,a) starts with a base distribution H, and a scaling parameter a.
Draw X_1 from distribution H.

for i > 1
With probability a/(a+i-1) draw X_i from H
With probability n_x/(a+i-1) set X_i = x, where n_x is the number of times x was chosen in the past

Polya's Urn

An urn contains a set of coloured balls of different colours. At each step, the following actions are performed:

- Retrieve a ball uniformly at random, from the urn.
- Replace the ball along with another ball of the same colour back to the urn.

Models nonlinear phenomena of opinions where publicly expressed opinions increase the proportion of the opinion in the population.

Ergodicity

Simple System = Linear (no feedback, closed loop systems, can be complicated but still simple and predictable)

Complex System = Non-linear (output given as feedback to input, open loop systems, unpredictable)
Social networks are complex systems.

Machines = Linear, Tractable (Controllable) / Predictable, Dynamics present but is by design.

Ergodicity represents an "irreducibility" property of certain stochastic systems, where the average behaviour over time of the system does not depend on the particular trajectory chosen initially.

A stochastic process on a population is said to be ergodic, if its statistical properties can be deduced from a single, sufficiently long, random sample of the process.

Ergodic hypothesis: In an ensemble of ergodic processes, the ensemble average at any time is equal to the (time) average of any process.

Example: Consider an ensemble of N Twitter accounts that post about a particular political party. Let μ_T be the average number of posts per day of any given account from this ensemble, when observed over a sufficiently long period of time. Let μ_N be the average number of posts from the ensemble on any given day. The system is ergodic if $\mu_T \cong \mu_N$

Beings = Unpredictable because of non-linear feedback but the system is quite stable because of invariant stable properties which makes such systems manageable, i.e. bounded state space with invariants. For example, human beings.

Chaos = Non-linear, Non-Ergodic, Intractable => may have no invariant/stable states.

Homogeneous Markov Chain

A 1-dimensional, infinitely long, discrete, finite-space random walk.
k-Markov assumption: Value of random variable at time t depends on up to k previous values.

Most commonly k=1 is used for stochastic models.

The probability distribution of the next state is dependent only on the current state X_t and can also be seen as the state of the Markov Chain at time t.

$$\forall t > 0, \forall I \subseteq \{0, 1, \ldots, t-1\}, \forall s_i, s_j, s_k \in S$$

$$Pr[X_{t+1} = s_j | X_t = s_i, \forall k \in I, X_k \ \ s_k] = Pr[X_{t+1} = s_j | X_t = s_i]$$

The random variable X_t denotes the state of the Markov chain at time t.
The term q_t (a vector) is used to denote the probability distribution of the Markov chain over the entire state space S.

The i^{th} element of q_t is given by,

$$q_t(i) = \Pr(X_t = i)$$

The Markov chain is said to be homogeneous if $\Pr(X_{t+1} = j | X_t = i)$ is independent of t. Finite homogeneous Markov chains can be represented as state transition systems with transition probabilities assigned to each transition, such that outgoing transition probabilities from each state sum up to 1.

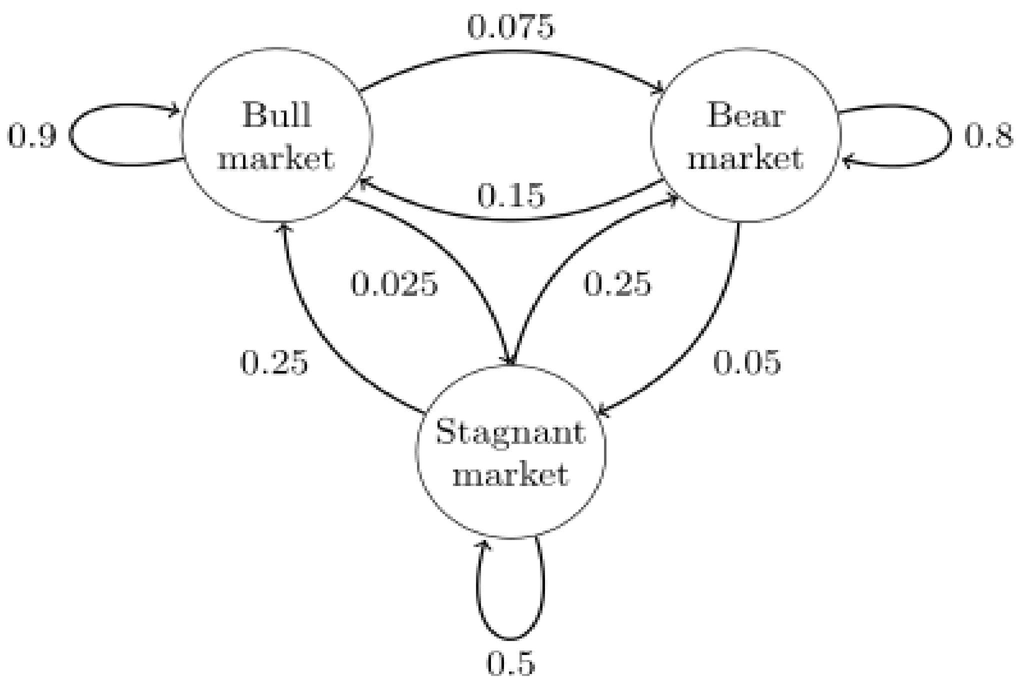
The adjacency matrix T of such a state machine is also called a transition matrix or a stochastic matrix.

Rows in the transition matrix T represent outgoing probabilities and sum up to 1.
Columns in the transition matrix represent incoming probabilities from other states.

The probability distribution for the entire state space at time t+1 can be computed as,

$$q_{t+1} = P \cdot q_t$$

Example,



You start with an initial probability distribution,

$$q_0 = [0.1 \quad 0.6 \quad 0.3]$$

These probabilities represent, Bear Market, Bull Market, Stagnant Market

And you have the transition probability matrix,

$$P = \begin{bmatrix} 0.9 & 0.075 & 0.025 \\ 0.15 & 0.8 & 0.05 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}$$

Now you can compute the state at time instant 1, by using the equation,

$$q_{t+1} = P \cdot q_t$$

Irreducibility: A Markov chain is said to be irreducible if for any k > 0 and for every pair of states, $i,\ j,\ \Pr(X_k = j | X_0 = s) > 0$. That is every state in the Markov chain is reachable from every other state with a non-zero total probability.

It is easy to see that a Markov chain is irreducible only if the state transition graph is strongly connected.

The period of any state s ϵ S of a Markov Chain is the GCD of all k > 0 such that, $\Pr(X_k = s | X_0 = s) > 0$. The Markov Chain is said to be aperiodic if all states have period 1.

Aperiodicity: From any state, you can come back to the same state in any number of steps.

A trivial graph that's both irreducible and aperiodic is a directed complete graph.

Stationary Probability Distribution

Given a homogeneous finite Markov chain with state space S and transition probability matrix T, a row vector over the set of states,

$$\pi = (\pi_s)_{s \in S}$$

is called a stationary probability distribution if,

$$\sum_{s \in S} \pi_s = 1 \ \ \& \ \ \pi \cdot T = \pi$$

Every irreducible finite Markov chain is shown to have a unique stationary distribution.
If a finite Markov chain is irreducible and aperiodic, the stationary distribution is independent of the initial distribution q_0. The Markov chain is said to eventually converge to this unique stationary distribution.

$$\lim_{t \to \infty} q_t = \pi$$

Interpretations of stationary distribution of a homogeneous Markov Chain:
Invariant property of influence
Principal eigenvector of adjacency

Aggregated distribution of traffic / attention / influence
PageRank and the stationary distribution

Interesting Discussion on Resource Consumption

Water Consumption is assumed to be per capita, but really it's not!
Consumption happens on a per community basis, i.e. institutions, malls, factories, etc., so if there's N people then there can be 2^N - N - 1 communities eliminating N singleton communities and the Null community.

So water consumption is not linear in terms of the population size and instead its exponential in terms of the population size.

So think of the world in terms of systems and not in terms of individuals, key takeaway. [Example, Rainwater harvesting doesn't solve water shortage problem in Bangalore for the long term because water consumption is exponential and not linear in terms of population]

END OF MANDATE 1

MANDATE 2: Structural Analysis of Networks

Structural Analysis of Networks

Analysis of connectivity patterns in a network for mining latent semantics.
Uses a graph representation of a network as the fundamental data structure.

Foundational question: How does the network position of an element (person, entity) affect its prospects?

Value of Network

Value of a network comes from the ability of the network to provide for the needs of any given agent.
"Network effect" or "network externality" used to denote the value provided for a given entity by virtue of it being part of the network.

Network effects:

Total effect (increase in value for all users of the network population, by virtue of forming the network).
Marginal effect (increase in value for any given individual, by virtue of participating in the network).

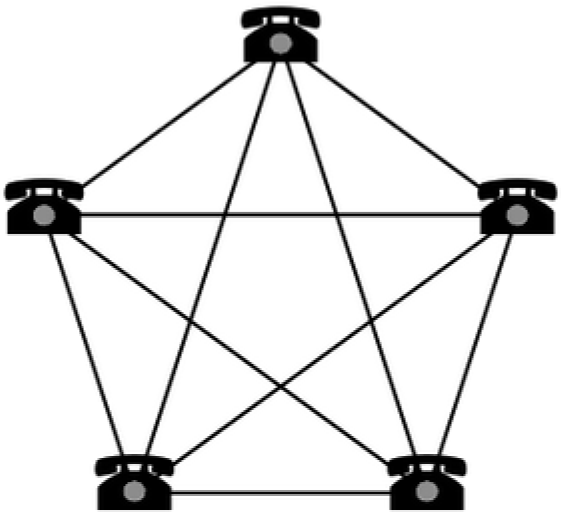
What does this mean?
Suppose a node in the network is an individual that is characterized by two things, their needs (associated with negative values) and talents (associated with positive values).
Example, I have good programming skills and I want someone to proofread my work.

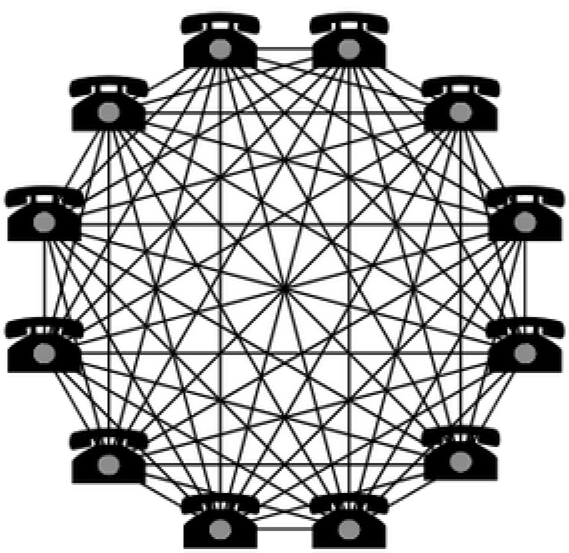
Nodes will form connections if they have a relation between their needs and wants, if a node has its need satisfied then the overall negative value of the node decreases but the person still has their talent so the positive value is still there.

Metcalfe's Law

The (total) value of a network given by number of possible pairwise connections among individuals in the network: $O(n^2)$

The (marginal) value of a network given by number of connections available for any given individual: $O(n)$





What does this mean? The value of having a telephone increases if other people in the network also have telephones.

Reed's Law

The total value of a network is the number of communities it can form from the population.
Number of possible communities from a population of n agents: $2^n - n - 1 = O(2^n)$

Negative Reed externality: Demand for resources (water, electricity, etc.) is proportional to the number of possible groupings of individuals in a population, and not linear to the number of individuals.

Marginal value of a network for an individual is the number of communities that an individual can be part of.

In a population of n individuals, number of communities (with at least one other individual) that an individual can participate:

2^{n-1}-n-2(derive it!)

If you remove yourself from the network, then you have n-1 people from which you can get, 2^{n-1} - n - 2 communities that you can participate in.

Beckstrom's Law

Valuation based on network dynamics.
Value of a network is defined as "net value of each user's transactions conducted through the network, summed up over all users' transactions".
Widely used in online networks like online shopping, mobile payments, etc.

$$\sum_{i=1}^{n} V_{i,j} = \sum_{i=1}^{n} \sum_{k=1}^{m} \frac{B_{i,j,k} - C_{i,j,k}}{(1 + r_k)^{t_k}}$$

$V_{i,j}$: value of network j to user i
$B_{i,j,k}$: payoff from transaction k to user i in network j
$C_{i,j,k}$: cost of transaction k to user i in network j
r_k : deprecation or discount rate for transaction k over time
t_k : time taken for completion of transaction k

The more the time taken, the lower the worth of the amount put in will be (can map this to time value of money and depreciation of currency).

Sarnoff's Law

Used in assessing broadcast media.
Value of a broadcast network proportional to its viewership size or number of viewers.

Extension to social media: Value of a social media network proportional to number of engaged users (on Twitter, engagement measured by number of replies, mentions and retweets).

Centrality Measures

A concept used to quantify the “importance” of a network element (node, edge, subgraph)
Like for connectivity, you can have node/edge connectivity.

Several measures of centrality exist, based on what is termed as “important”.

Based on problems in several areas: influencer identification, facility location, traffic modeling, stress mapping, etc.

Degree Centrality

Given a finite, connected, undirected graph
G = (V,E)
The degree centrality of any node is the proportion of its degree to the total degree in the graph.

Degree centrality is a measure of the extent to which a node can directly affect (influence) other nodes in the graph.

So higher the degree of a node, the higher its importance in the network.

Formally,

$$c_D(v) = \frac{d(v)}{\sum_{\forall v \in V} d(v)}$$

where d(v) is the degree of node v.

Degree centrality of any node in a regular graph is 1/n.

Eccentricity

Eccentricity is a centrality measure in facility location problems.

Some facility location tasks (**locating emergency services** like hospitals) require one to minimize the maximum distance from any node in the network.

Eccentricity of a node v is the maximum distance from the node v to any node in the network:

$$e(v) = \max_u(d(v,u))$$

Time Complexity to compute this,
$O(n^2)$ computing the distance between every pair of nodes.
So in the example of facility location, all locations/houses/etc. would be the nodes in the network and our problem is to place hospitals at these nodes such that the distance to a hospital from all other nodes is minimum.

Centrality of a node v based on its eccentricity is defined as:

$$c_E(v) = \frac{1}{e(v)}$$

Nodes with minimum eccentricity are called “graph centers”.
And this minimum eccentricity is called the graph radius.

A free tree has at most two centers. An arbitrary graph has any number of centers.

Why does a free tree have at most two centers?
Keep removing the dangling nodes (nodes with degree 1) from the graph till you cannot remove any dangling node, there can be at most one or two nodes remaining, i.e. the center.

Radius
Given a connected, undirected graph G = (V,E) its radius is defined as the eccentricity of the least eccentric node. In other words:

$$r(G) = \min_{\forall u \in V} e(u)$$

Time Complexity to compute this is the same as Eccentricity.

The center of a graph based on its radius is given by:

$$C(G) = \{u \in V,\ r(G) = e(u)\}$$

What is an induced subgraph?
Get a subset of vertices and the edges that connect between these vertices from the original graph. This would be the induced subgraph.

If you take only the graph centers and the paths between these graph centers as a subgraph (not an induced one), you’ll find that the connectivity of this subgraph is atleast 2 (proof not covered) which means that there’s atleast 2 paths from one graph center to another, i.e. you can quickly move from one graph center to another and from there you can move to the entire graph with minimal distance.

14/10/22

Closeness Centrality

The closeness or minisum measure of a node v, depicted as c(v), is the sum of its distances from all other nodes in the network.

$$c(v) = \sum_{u \in V} d(v, u)$$

Centrality based on closeness is defined as:

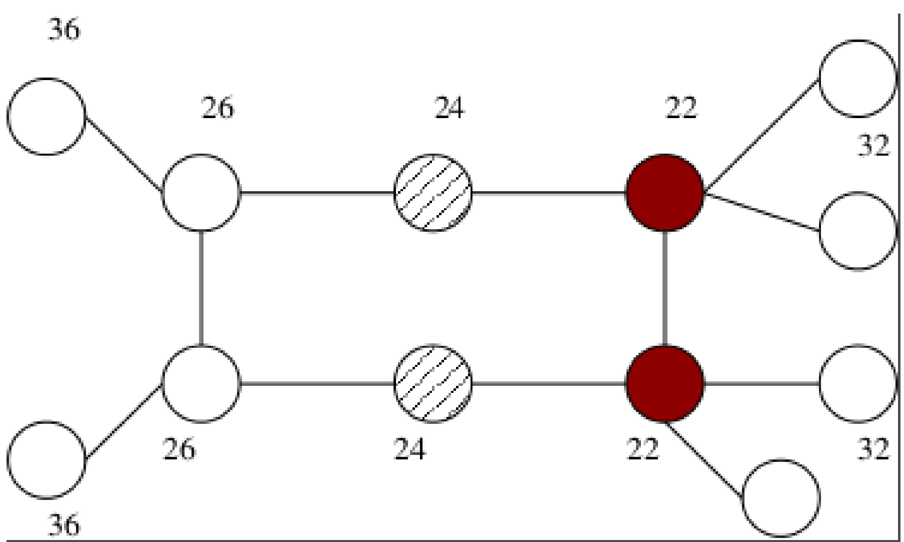
$$c_C(v) = \frac{1}{c(v)}$$

Lower the closeness of a node, i.e. lesser the sum of distance of the node from all other nodes, higher is the closeness centrality and so that node is generally said to be “closer” to the other graph nodes.

Closeness centrality pertains to facility location **for a utilitarian entity** (like a shopping mall) maximizing access to as many customers (nodes) as possible.

Nodes with highest closeness centrality, also called graph medians.

Filled nodes represent central nodes w.r.t. closeness, while hatched nodes represent central nodes w.r.t. eccentricity.



Graph Medians
Graph medians are defined based on the closeness centrality measure.

Given a graph G = (V,E), the minimum total distance is given by

$$s(G) = \min_{\forall u \in V} c(u)$$

The median M(G) is given by,

$$M(G) = \{u \in V : \; s(G) = c(u)\}$$

Properties of graph centers based on radius (see the two theorems from earlier) also hold for graph medians.

Graph centers and graph medians need not be the same and they could be separated by arbitrarily long distances.

Radiality

The radiality measure of a node is a measure of how ``closely integrated'' is the node in the network.

Given a graph G with n nodes, if Δ_G denotes the diameter of the graph, the radiality of a node v is given by:

$$r(v) = \sum_{u \in V} \Delta_G + 1 - d(v, u)$$

Centrality of a node w.r.t. its radiality is its radiality normalized over all nodes:

$$c_R(v) = \frac{r(v)}{n-1}$$

Radiality measures are useful for locating land transport hubs like the central railway station of a town.

Normalization w.r.t the diameter makes the radiality score comparable against the network edges.

Example: “10 kms to the central railway station -- is it too far or very near?”
Distances are usually relative and in terms of viewing the entire system as a graph, distances are relative to the diameter of the graph. And this is what the Radiality measure deals with.

Higher radiality means that the node is connected to a lot of nodes with relatively shorter distance. Lower radiality means the node is connected to very few nodes with relatively longer distance.

Also note that distances are still subjective depending on the mode of transport, like transport by car, bus, walking, etc. all have different status associated with fixed quantity of distance like 10km, for bus it’s a bit much, for car it’s not too much, for walking it’s a lot.

Eccentricity => For emergency point of view.

Closeness Centrality => For utilitarian (profit-making) point of view.

Radiality => Has both utilitarian (profit-making) point of view but also an emergency point of view.

Sidenote:
UPI industry reduced the sales of the toffee industry significantly because there's no need for change anymore.
ARPU = Average Revenue Per User

Traversal Based Centrality Measures

Pertains to computing the importance of a graph element (node or edge) based on its participation in shortest paths in the graph.
Primary linked to traffic modeling (assume that traffic flows through the shortest path).

Measures various factors like the stress on the graph element, the influencing power of the graph element, the vitality of the element, etc.

Example centrality elements:
Stress centrality
Betweenness

Stress Centrality

Assuming that traffic in a network takes the shortest route to go from source to destination, the number of shortest paths that a graph element (node or edge) v is involved in, represents the amount of "stress" on the element.

The assumption of shortest route is dependent on many factors like time (how long travel takes), distance (how much does one have to travel), capacity (how many people can travel at the same time on the route), etc.

The value of a route being short is also dependent on transportation factors like a node is participating in many shortest paths, because that node has metro connectivity to many other nodes which will take very less time and has good connectivity (example, Silk Board).

Let $\sigma_{st}(v)$ denote the number of shortest paths between s and t that contains element v. Given this, the stress centrality of a node v is:

$$c_S(v) = \sum_{s \neq v} \sum_{t \neq v} \sigma_{st}(v)$$

Analogously, the stress centrality of an edge e is:

$$c_S(e) = \sum_{s \in V} \sum_{t \in V} \sigma_{st}(e)$$

Time Complexity to compute this is,
$O(n^2 * n^2 * n) = O(n^5)$
For each vertex (n nodes), for every pair of vertices (n^2 combinations), compute the shortest paths between them (n^2 computations), so we get n^5 in total.

Betweenness Centrality

Given two nodes s and t, the betweenness of a graph element (node or edge) v is the ratio of the number of shortest paths between s and t and those that pass through v:

$$\delta_{st}(v) = \frac{\sigma_{st}(v)}{\sigma_{st}}$$

The betweenness centrality of graph element v is its total betweenness for all pairs of nodes in the graph:

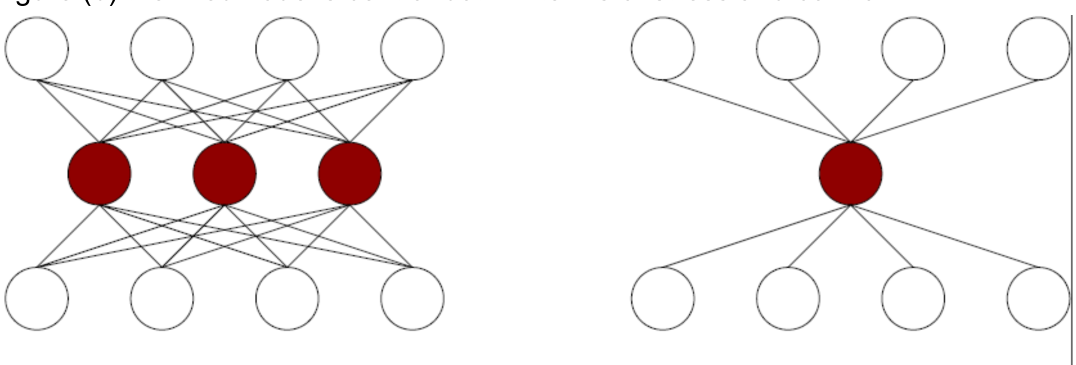
$$c_B(v) = \sum_{s \neq v} \sum_{t \neq v} \delta_{st}(v)$$

So betweenness deals with the stress of a node's participation in the shortest paths but also deals with the exclusivity of a node's participation in the shortest paths.

Betweenness vs Stress

Betweenness measures the relative influence or control that a graph element has, while stress centrality measures the relative load that the graph element takes w.r.t. other elements.

In figure (a), filled nodes have high stress centrality but low betweenness centrality; while in figure (b) the filled node is central both in terms of stress and control.



(a) (b)

Generalized Betweenness

Why do we need generalization?
Because assumptions may not always hold in the real world.

Betweenness $P_{st}(v)$ can be calculated over any set of paths P_{st} (not just shortest paths) across nodes s and t.

A good alternative is to take the k-shortest paths instead of the shortest paths between pairs of nodes.

There's many other generalized versions of Betweenness but the conclusion is that **in practical scenarios, base centrality measures usually don't amount to much and assumptions won't exactly hold.**

Vitality

Let G = (V, E) be a graph and let f: G->R be a real-valued function on G that represents some property of G. Then for any graph element e (node or edge), its vitality is defined as the change in the value of f when e is removed.

Vitality(G) = f(G-{v}) - f(G)

Some examples for f():
Diameter
Connectivity
Number of centers
Number of medians

Vitality for Free Tree with f() as Connectivity
Free Tree Connectivity = 1
Remove any vertex/edge, Connectivity is now 0
Therefore, Vitality = 0 - 1 = -1

Flow Vitality

Let G = (V, E, w) be a directed, weighted graph with non-negative weights on edges denoting their capacity. Given two nodes s, t ∈ V, the maximum flow between them is equivalent to the minimum capacity of an s-t-cut (max flow-min cut, i.e. Ford-Fulkerson Theorem).

Let $f_{s,t}$ denote the capacity of the maximum flow between nodes s and t (that is, minimum flow capacity amongst all the paths between s and t) and let $f_{s,t-u}$ denote the capacity of the maximum flow between s and t with node u removed from the network.

The flow vitality of u is then given by:

$$c_{fv}(u) = \sum_{u \neq s, u \neq t, f_{st} > 0} 1 - \frac{{f_{st}}^{-u}}{f_{st}}$$

Measures how a node affects the max flow of the network.

Wiener Index 🌭🍆

Given a graph G = (V,E) the Wiener Index is defined as the sum of all distances over all vertex pairs:

$$I_W(G) = \sum_{u \in V} \sum_{v \in V} d(u,v)$$

The Wiener index can also be seen as the sum of the closeness centrality values of all vertices in the graph:

$$I_W(G) = \sum_{v \in V} \frac{1}{c_C v}$$

Closeness Vitality

Based on the Wiener index, the closeness vitality of a graph element (node or edge) can be defined as:

$$c_{CV}(x) = I_W(G) - I_W(G - \{x\})$$

Closeness vitality denotes the extent to which the cost of all-pairs communication increases with the removal of the element -- denoting the vitality of the element for the network

If x is a cut node or edge, then the closeness vitality becomes -∞.

Shortcut Vitality

Given any graph G = (V,E), and an edge e = (u,v), the shortcut vitality of edge e is defined as the maximum increase in distance between any two vertices if e is removed from the graph.

The only pairs of nodes affected would be those which have e in the shortest path between them.

Maximum increase in distance will be for the pair (u,v) itself.

If e is a cut edge, the shortcut vitality score will be ∞

Stress Vitality

Given a graph G = (V, E) and an element x, the stress vitality of x is the number of shortest paths that are lost in the graph with the removal of x.

Note that between any two nodes in a graph, there may be several shortest paths and some are lost with the removal of an intermediate element (node or edge).

Highly linked to Braess's Paradox, increasing the size of the road doesn't decrease the traffic because the max flow between any two destination points going through the large road is still the same and the stress on the connecting roads (smaller capacity ones) to the larger road increases.

Formal definition of stress vitality:

Given a graph G = (V,E) and an element x, let the term f(G-{x}) be defined as:

$$f(G - \{x\}) = \sum_{s \in V} \sum_{t \in V} \sigma_{st}[d_G(s,t) = d_{G-\{x\}}(s,t)]$$

That is, f(G-{x}) is the sum of all shortest path distances that have remained the same before and after removal of element x.

Given this, the stress vitality of element x is:

$$c_{sv}(x) = f(G) - f(G - \{x\})$$

Web Centralities

Centrality measures developed in the context of hyperlink graphs on the web.
Based on cognitive and social notions like attention, authority, citizenship, etc.

Early web centrality measures: PageRank, HITS

PageRank

Simplest model of the behaviour of a random surfer on the web – as a random walk on the hyperlink graph.

Given a hyperlink graph G = (V,E) depicting web pages and hyperlinks, the page transition matrix T is given by,

$$t_{ij} = \begin{cases} \frac{1}{d^+(i)} & (i,j) \in E \\ \frac{1}{n} & d^+(i) = 0 \end{cases}$$

where, $d^+(i)$ = Outdegree of node i

Note that a "sink" page with no outgoing links is said to connect to all pages with equal probability.

The PageRank centrality of page p is computed based on the number of pages pointing to p and their PageRank. This kind of recursive formulation is also called a **feedback centrality measure.**

$$c_{PR}(p) = d \sum_{q \in \Gamma_p^-} \frac{c_{PR}(q)}{d^+(q)} + (1-d)$$

$where,\ \Gamma_P^- = \text{Nodes that have incoming edge to p}$
$d = \text{Dampening Factor},\ 0 \le d \le 1$

Correspondingly,
$C_{PR} = d \cdot T \cdot C_{PR} + (1-d) \cdot 1_n$

Note that this model of the web gives us three properties, Homogeneity, Irreducibility and Aperiodicity because of which we get a Stationary Probability Distribution (PageRank) which is independent of the initial probability distribution.

HITS

An algorithm based on the notion of "hubs" and "authorities" where:

A good hub is a page that points to many good authorities.
A good authority is a page that is pointed to by many good hubs.

Each node in a hyperlink graph is associated with two scores: a hub score and an authority score. If A is the adjacency matrix, the respective scores are computed as:

$$\vec{a} = A \cdot \vec{h}$$

$$\vec{h} = A^T \cdot \vec{a}$$

Since neither values are known, recursive formulation solved using power iterations:

$$\vec{a}^0 = \mathbf{1}_n$$

$$\vec{a}^{i+1} = A \cdot \vec{h}^i$$

$$\vec{h}^{i+1} = A^T \cdot \vec{a}^i$$

Properties of Hub & Authority Scores

The vector of authority scores $\vec{h}$ corresponds to the principal eigenvector of the matrix A^TA.
The vector of hub scores $\vec{h}$ corresponds to the principal eigenvector of the matrix AA^T.

Affiliation Network

Bipartite graph. Good model of where HITS algorithm is useful.
Example, People on the L part of the graph and Institutions on the R part of the graph.
So an edge in this Bipartite graph is an affiliation of the person node to the institution node.

PageRank vs HITS

A node with high PageRank diffuses its PageRank to its outdegree nodes in a depth-first manner.
Example, a Page got a boost in search results increasing its PageRank, the pages linked in that page get a boost and the pages that are linked in these pages get a smaller boost and so on, and this is the depth first nature of the diffusion.

A node with high Affiliation score (HITS scores) diffuses their score laterally, Hub score diffuses to Authority scores of its neighbours and Authority scores diffuse to Hub scores of their neighbours.
Example, Company becomes very valuable and famous -> Founders become popular -> Other companies where founders are present on the board become popular. This is lateral diffusion.

HITS is not that useful on the Web because it factors in outdegree which can be easily manipulated on the Web, but it is quite useful in affiliation networks (founder and company network for example) where links cannot be easily manipulated.

So HITS is more robust because it considers both indegree and outdegree which is also its problem when it comes to the Web which is why PageRank is preferred in that case.

PageRank -> Diffuses Attention in depth-first manner
HITS -> Laterally diffuses Brand Value

Agent Based Centrality Measures

These measures are based on modeling some kind of an interaction between nodes in a network.

Each node in the network is associated with a “value” or “score”, s (also called “self interest” or “utility”), which it optimizes for itself based on its interactions.

For example, the network nodes are people and the edges represent relations between people, i.e. friends, relatives, online friends, etc.
Now a belief is measured across this network, for example, “Are iPhones expensive?”.
The belief is modelled as a value between 0 and 1, 0 = Strongly disagree, 1 = Strongly agree.

If you are at a belief of 0.5 and your neighbour nodes have polar opposite beliefs, i.e. 0 and 1, then there’s a chance for instability in the network due to which connections can be severed :(
Another option is to revise your belief to reflect your connections better and this is what these centrality measures generally cover.

The network is said to have reached a stable state or equilibrium, when each node has no incentive to change its values.

Useful in modeling several kinds of social interaction based centrality measures.

Feedback based centralities are typically defined on real-valued, weighted, undirected, loop-free graphs.

Bonacich Centrality

Given an undirected social network, G = (V, E, w) with real-valued weights w on edges E representing the strength of acquaintance between pairs of nodes.

Every node i is assigned an initial “sociability” score s_i

Sociability scores are adjusted based on minimizing the disparity between sociability scores of adjacent nodes, and their acquaintance weights.

Specifically, the following measure (Sociability score) is minimized across the network, with an initial assignment of s_i = 1 for all i:

$$\min_{s_i, s_j} \sum_{\forall i,j \in V} (s_i s_j - a_{i,j})^2$$

$a_{i,j}$ = Weight between i and j

The sociability score indicates the optimal belief value that a person should have towards a belief so that the stress in the network is minimized, stress as in maintaining the connections that the person has, even though their beliefs might be different.

21/10/22

Hubbell Centrality

Similar to Bonacich centrality, but with some important differences.

Sociability score of a node, measured as a fixed-point computation (diffusion process), rather than as an optimization process.

Sociability score of each node computed as a weighted sum of sociability scores of all its adjacent nodes.

Directly extendable to directed, weighted graphs.

$$s_i^{k+1} = \sum_{j \in \Gamma(i)} a_{i,j} \cdot s_j^k$$

$a_{i,j}$ = Weight between i and j
k = Iteration number

In Bonacich Centrality, each node has a belief (modelled as the sociability score) and nodes try to minimize the score on the basis of sociability of its neighbours.

In Hubbell Centrality, each node has a belief that it diffuses to its outdegree neighbours and this process happens in iterations in depth-first manner, similar to PageRank.

Belief propagation happens in the reverse direction of the edges, i.e. based on who I follow, my beliefs are influenced and based on who follows me, their beliefs get influenced.

General behaviour observed in belief and opinion networks.

Estimating Centrality Measures

Many centrality measures are hard, or even intractable, to compute on large graphs.

In many cases, a good estimation of the centrality measure of a given graph element is sufficient.

Estimation of centrality scores based on graph sampling -- creating a smaller sample graph (or a set of sample graphs) such that they retain the structural properties of the larger graph.

Random walks, a critical component of creating graph samples.

Degree Centrality

Given a simple undirected graph G = (V, E), the stationary probability of a canonical random walk on the graph is the distribution of the degree centrality of the graph.

Proof idea: Begin with the following, the transition probability matrix of a random walk P is given by,

$$p_{ij} = \frac{a_{ij}}{d(i)}$$

The stationary distribution of a random walk π

Proof idea: Begin with the following – the transition probability matrix of a random walk P is given by

$$p_{ij} = \frac{a_{ij}}{d(i)}$$

The stationary distribution of a random walk π is given by: $\pi \cdot P = \pi$
Now show that $\pi_i = \frac{d(i)}{\sum_{v \in V} d(v)}$ over P is stationary.

Betweenness

Let G = (V,E) be a simple, finite, undirected, unweighted, connected graph.

Let s,t be two vertices of G such that s wants to send a message to t, but does not know connectivity information about the graph.

The simplest way to model discovery of a path from a source to destination given local information, is to assume that each node forwards the message to its adjacent nodes uniformly at random. Every node, except the destination t forwards the message

Let,

$$m_{i,j} = \begin{cases} \frac{a_{i,j}}{d(j)} & if j \neq i \\ 0 & otherwise \end{cases}$$

denote the probability that a message is received by node i from node j.

Computing shortest paths is quadratically hard atleast, so instead of that we start some random walks from the source and terminate a walk when it reaches the destination.

The idea here is that the random walker on the shortest path is most likely to reach the destination node the fastest. This overcomes the effort of computing the shortest paths between both nodes and returning a reasonably short path between them.

Let M be the matrix denoting all receipt probabilities in a given time step and let D be a diagonal matrix denoting the degrees of each vertex. The inverse of the degree matrix D^{-1} is also a diagonal matrix containing the reciprocal or inverted vertex degree for each vertex. Let A be the adjacency matrix of the graph.

We can see that

$$M_{-t} = A_{-t} \cdot D_{-t}^{-1}$$

where $M_{-t}, A_{-t}, D_{-t}^{-1}$ denotes the corresponding matrices with the row and column t removed.

The r^{th} power of M_{-t}, denoted as M_{-t}^{r} contains the probability for each node i receiving a message sent out by each other node j in r steps

Note that M_{-t}^{∞} exists, since all values of M are between 0 and 1. M_{-t}^{∞} denotes the random walk betweenness of each node for destination t.

Closeness

An approximation to closeness centrality can be obtained by a random-walk related measure: Mean First Passage Time (MFPT)

The MFPT $m_{st,}$ is defined as the expected number of nodes, a message starting at node s has to travel before it encounters t for the first time.

We start random walks from many nodes in the sample of the graph while computing the closeness for the graph.

How many steps did the random walker take before visiting v for the first time?
Average of this for all random walkers is the MFPT which is a good estimate for the closeness centrality.

$$m_{st} = \sum_{n=1}^{\infty} n \cdot f_{st}^{(n)}$$

where $f_{st}^{(n)}$ is the probability that t is visited for the first time after exactly n steps.

The Markov closeness centrality for a node v is defined as the inverse of the average MFPT for all random walks starting in any node s and ending at node v.

$$c_M(v) = \frac{n}{\sum_{s \in V} m_{sv}}$$

Groups and Densities

Formation of groups or clusters is a fundamental characteristic of any social network.

Understanding groups can reveal several hidden semantics. Formation of groups may be based on several criteria like:

- Mutuality (group members are acquainted to one another, Triadic closures come to mind)
- Compactness (group members are highly reachable from one another)
- Density (group members have high closure w.r.t. adjacency)

- Clustering (group members are more connected within the group than outside)

Cliques

Also called “Perfectly dense groups”.

Given a simple, undirected graph, G = (V, E), a subset of the nodes U $\subseteq$ V is called a clique if G(U) is a complete graph. Cliques are called perfectly dense groups and have the following properties,

- A clique of size k has a minimum degree of k-1
- It has a diameter of 1
- It is perfectly connected, i.e. it has k-1 vertex and edge connectivity

A clique U is said to be maximal if there is no U $\subset$ U’ which is also a clique.

More properties of Cliques
- Cliques are closed under exclusion. That is, if U is a clique and v is a vertex in the clique, then U - {v} is also a clique.
- Cliques are nested. Every clique of size n contains a clique of size n-1 and so on.

Plex

A “plex” is a generalization of the concept of a clique, where nodes are allowed to “miss” having edges with other nodes.

Let $G = (V, E)$ be an undirected graph with $|V| = n$, and let $1 \leq N \leq n-1$ be a natural number. A subset $U \subseteq V$ is said to be an N-plex iff $\delta(G[U]) \geq |U| - N$

$\delta(G[U])$ is the minimum degree of any node in graph U.

A 1-plex is also called a *clique*

Properties
- Plexes are closed under exclusion. Any subgraph of an N-plex is also an N-plex.
- Relationship between an N-plex and diameter and connectivity:
An N-plex on a graph with n nodes, with n >= 4 and N < (n+2) / 2 has an edge connectivity of atleast 2.
An N-plex with N >= (n+2)/2 has diameter no more than 2N - n + 2

Detection of maximal cliques or N-plexes are known to be NP-complete

Cores

A **tractable notion** of a dense subgroup.

Based on the minimum degree in a subgraph.

Given a graph G = (V,E) a subset of nodes U is said to be an N-core iff minimum degree in U is at least N

U is said to be a maximal N-core if it is not contained in any larger N-core within G

An (N+1)-core is also an N-core and any N-core is an (n-N)-plex.

Cores are more tractable than Plexes because Cores can be more easily computed. How?
Find out the degree of each node and sort them by the degrees.
For each node with degree k, look at all its neighbours and if any neighbour has lesser degree, then remove it from the core. Similarly repeat this in a breadth-first manner.
This is an $O(n^2)$ algorithm.

Spectral Graph Segmentation

Consider an undirected graph G = (V,E) with its adjacency matrix represented by A and degree matrix as D.
We have the Laplacian matrix for the graph as:

$$L = D - A$$

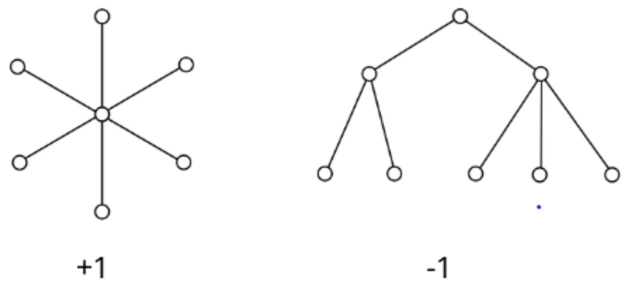
The Laplacian matrix has the following properties:
- Any row or column sums to 0.

$$\sum_j L_{i,j} = 0$$

$$\sum_i L_{i,j} = 0$$

0 is the most significant eigenvalue with $\vec{1}$ as the eigenvector

Consider that the graph is partitioned into two components.
Now create a vector of the form $s_i \in \{+1, -1\}$ where a node is given a value of either +1 or -1 depending on which component it belongs to.



We can see that this vector is also an eigenvector of the Laplacian matrix with eigenvalue 0. Prove this!

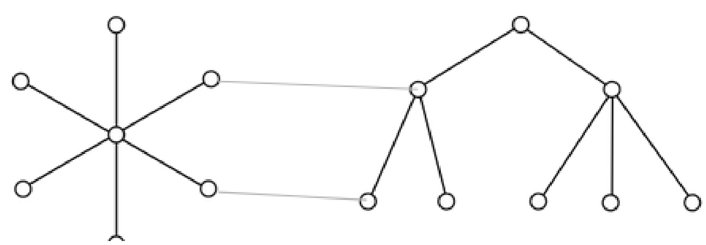
Simple hint, the Laplacian measures the relative standing of a node with respect to its neighbours and here all nodes are at the same level (+1 or -1) when compared to its neighbours. So the graph derivative still comes out to be zero.

Say we want to partition a graph, in this case, we try to minimize the graph derivative for each node in an iterative manner (using an algorithm like gradient descent) where the x vector can take a fixed set of values like {1, 2}, etc.

Eventually the graph derivative will stop decreasing and start increasing at which point we will have reached a local minima and this is a vertex where we can cut the graph.

This is spectral graph segmentation and we can create more clusters by repeating this iterative process further.

Suppose that instead of being partitioned, the components were connected by a small set of links.



This would make the product: L * s close to zero but not exactly zero.
Spectral graph partitioning is essentially assigning a value $s_i \in \{+1, -1\}$ for every node i under the following minimization constraint:

$$\arg\min_{s} \frac{\vec{s}^T L \vec{s}}{\vec{s}^T D \vec{s}}$$

Interpretation of this optimization value: removing which edge (or a cut) has the least effect on the flow of values (x) in the graph.

Normalized Cuts Algorithm

Let G = (V, E, w) be an undirected weighted graph and let A and B be two subsets of vertices. Define,

$$w(A, B) = \sum_{i \in A, j \in B} w_{i,j}$$

Let,

$$ncut(A, B) = \frac{w(A,B)}{w(A,V)} + \frac{w(A,B)}{w(B,V)}$$

represent the weight of the cut between A and B or its “conductance”.

Graph segmentation involves finding a subset of vertices S such that,

$$\arg\min_{s} ncut(S, \bar{S})$$

ncut Algorithm

Given an undirected graph G = (V, E, w)

Solve the equation:

$$\vec{s}^T L \vec{s} = \vec{s}^T D \vec{s}$$

for the second smallest eigenvector.

Partition the eigenvector s into two components based on its median component.
Recursively segment the components to create more segments.

Modularity

A measure of density of a subgroup of nodes in a graph based on a comparison with its density due to random chance.

Let G = (V,E) be a (directed or undirected) possible multi-graph, with adjacency matrix A. Let the total number of edges in the graph be m.

Between any distinct nodes i and j with degrees k_i and k_j, the expected number of edges is given by:
Degree of node 1 * Degree of node 2 / (2 * total degree of the graph)

$$\frac{k_i k_j}{2m}$$

m = Total number of edges = Total degree of graph

The “modularity” of the graph is then computed as:

$$\frac{1}{2m}\sum_{i,j\in V}\left(A_{ij}-\frac{k_i k_j}{2m}\right)$$

This measure basically tells us whether for each edge, whether the presence/absence of an edge matches with our expectation.

Measure is 0 when adjacency perfectly matches our expectation.
Measure is -ve when edge is absent and expectation is non-zero, it is +ve when edge is present and expectation is non-zero.

Suppose the graph is partitioned into two classes or communities named 1 and -1. Let the term s_i = 1 denote that vertex i belongs to community 1, same for -1 community.

The modularity of the partitioned graph is simply the modularity score summed over all pairs of vertices. This is multiplied by the term:

$$\frac{s_i s_j + 1}{2}$$

which reduces to 1 if both i and j belong to the same community, or 0 otherwise. Hence the overall modularity of a partitioning is given by:

$$Q_s = \frac{1}{2m}\sum_{i,j\in E}\left(A_{ij}-\frac{k_i k_j}{2m}\right)\frac{s_i s_j + 1}{2}$$

A "good" partitioning of a graph into two communities is one that maximizes the overall modularity score:

$$\arg\max_s Q_s$$

This is known as the Girvan-Newman Algorithm for clustering a graph.

Quiz on 4th November

END OF MANDATE 2